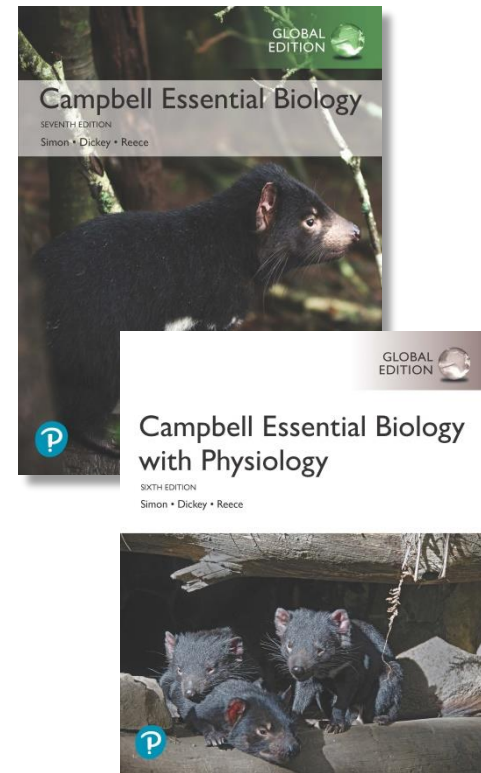


Campbell Essential Biology, Seventh Edition, Global Edition and Campbell Essential Biology with Physiology, Sixth Edition, Global Edition

Chapter 27 Nervous, Sensory, and Locomotor Systems



PowerPoint® Lectures created by Edward J. Zalisko, Eric J. Simon, Jean L. Dickey, and Jane B. Reece

Heads Up! Sports-related Head Injuries Can Cause Neurological Disorders



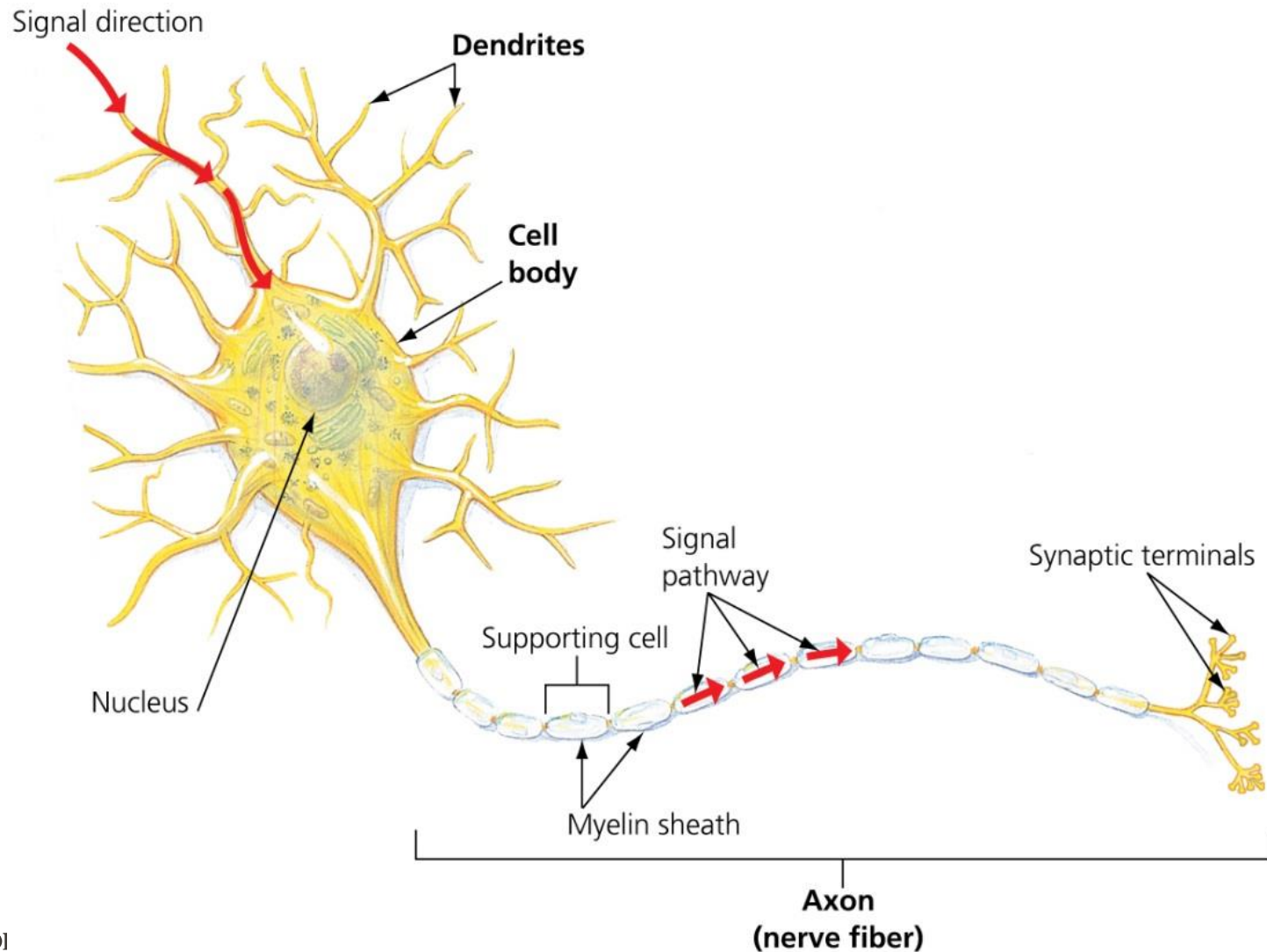
Biology and Society: Medicinal Poisons

- Deadly **neurotoxins**, produced by plants, animals, and bacteria can be **powerful tools in medicine**.
 - When medicinally administered in low doses, neurotoxins can benefit humans.
 - Researchers are engineering breakthrough treatments for chronic pain, seizures, and muscular dystrophy based on natural toxins.
 - Toxic animals and plants helped researchers discover some of what we know about the nervous system, your body's communication and coordination network.

An Overview of Animal Nervous Systems: Neurons

- The **nervous system** forms a communication and coordination network throughout an animal's body.
- **Neurons** are nerve cells that carry electrical signals from one part of the body to another.
 - Each neuron may communicate with thousands of others, forming networks that enable us to move, perceive our surroundings, learn, and remember.
 - Neurons can vary in structure, but most have structures to receive and send signals.

Structure of a neuron



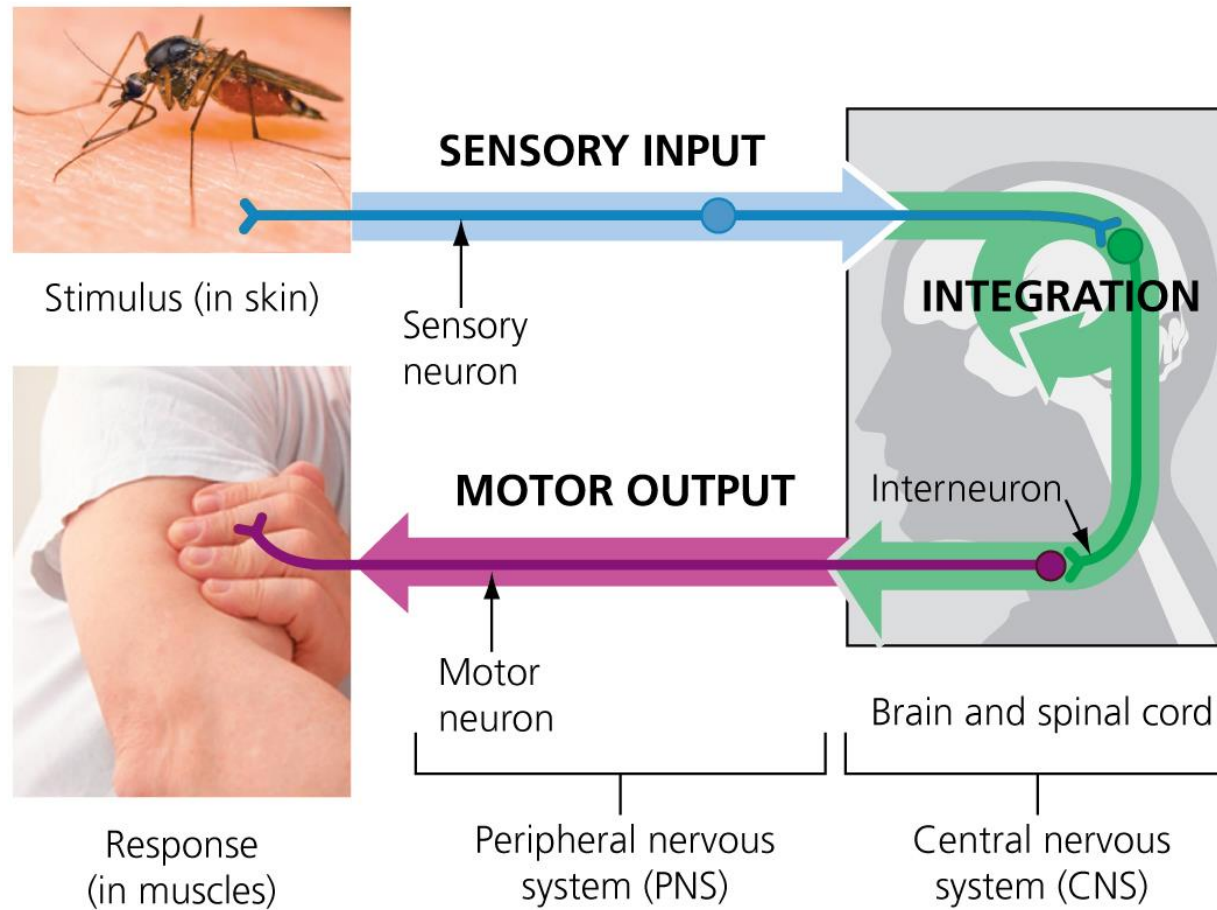
Neurons

- Many short and highly branched extensions, called **dendrites**, receive messages from other neurons and convey this information toward the cell body, which houses the nucleus and other organelles.
- A different type of extension, a single long fiber called an **axon**, or nerve fiber, transmits signals toward a receiving cell, such as a muscle cell.
 - A typical axon has hundreds or thousands of branches. Each branch ends with a **synaptic terminal**, which relays signals to a receiving cell.
 - The speed of nerve signals can be attributed to the **myelin sheath**, a chain of bead-like supporting cells that wrap around and insulate nerves.

Organization of Nervous Systems (1 of 2)

- The nervous systems of most animals have two main anatomical divisions.
 1. The **central nervous system (CNS)** consists of the brain and spinal cord and is the control center of the nervous system, interpreting stimuli, and sending out responses.
 2. The **peripheral nervous system (PNS)** is made up mostly of **nerves** (bundles of axons) that carry signals into and out of the CNS.
- **Figure 27.2** highlights the three interconnected functions of the nervous system and the three types of neurons that carry out these functions.

Organization of a nervous system



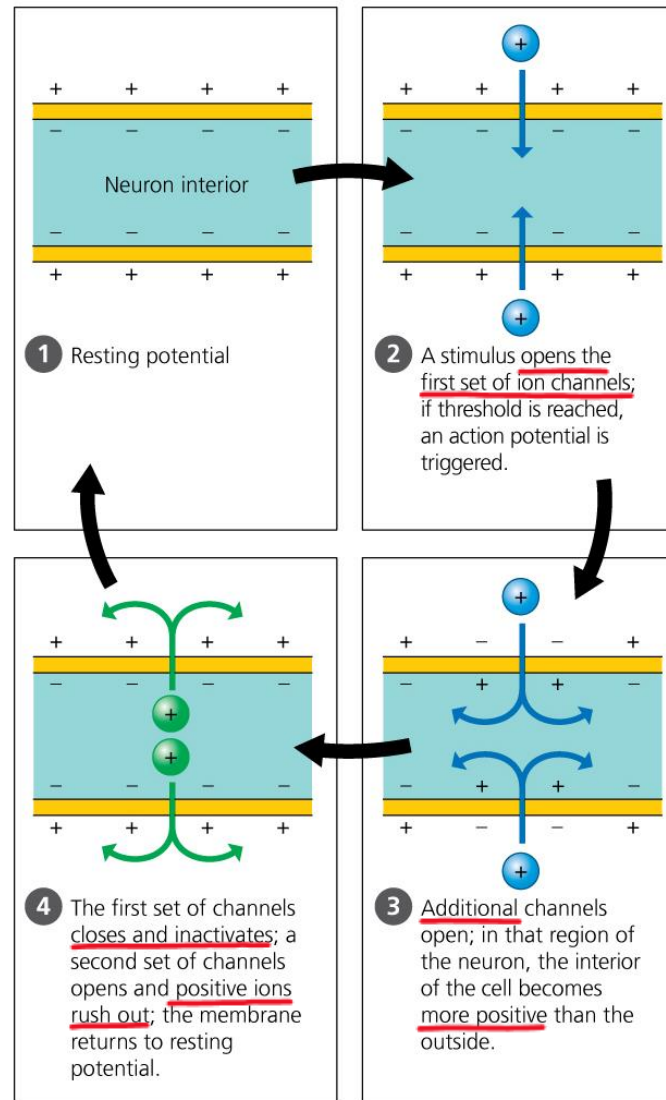
Organization of Nervous Systems (2 of 2)

1. **Sensory neurons** are nerve cells often located in the PNS that function in **sensory input**, sending information about a stimulus to the CNS.
2. During **integration**, sensory signals are interpreted and appropriate responses are planned. The nerve cells that integrate the sensory input, **interneurons**, are typically located in the CNS. Most neurons in the brain are interneurons.
3. **Motor neurons** function in **motor output** sending out signals from the CNS to the PNS that result in the body's response. Motor neurons carry output signals to muscles and glands.

Sending a Signal through a Neuron: The Action Potential

- At rest, a neuron's plasma membrane has a **resting potential** caused by the membrane's ability to maintain a positive charge on its outer surface that opposes a negative charge on its inner surface.
 - A stimulus alters a portion of the membrane, allowing ions to enter and exit the neuron. This creates an **action potential**, a signal that carries information along a neuron.
 - **Figure 27.3** shows the series of events involved in generating an action potential.

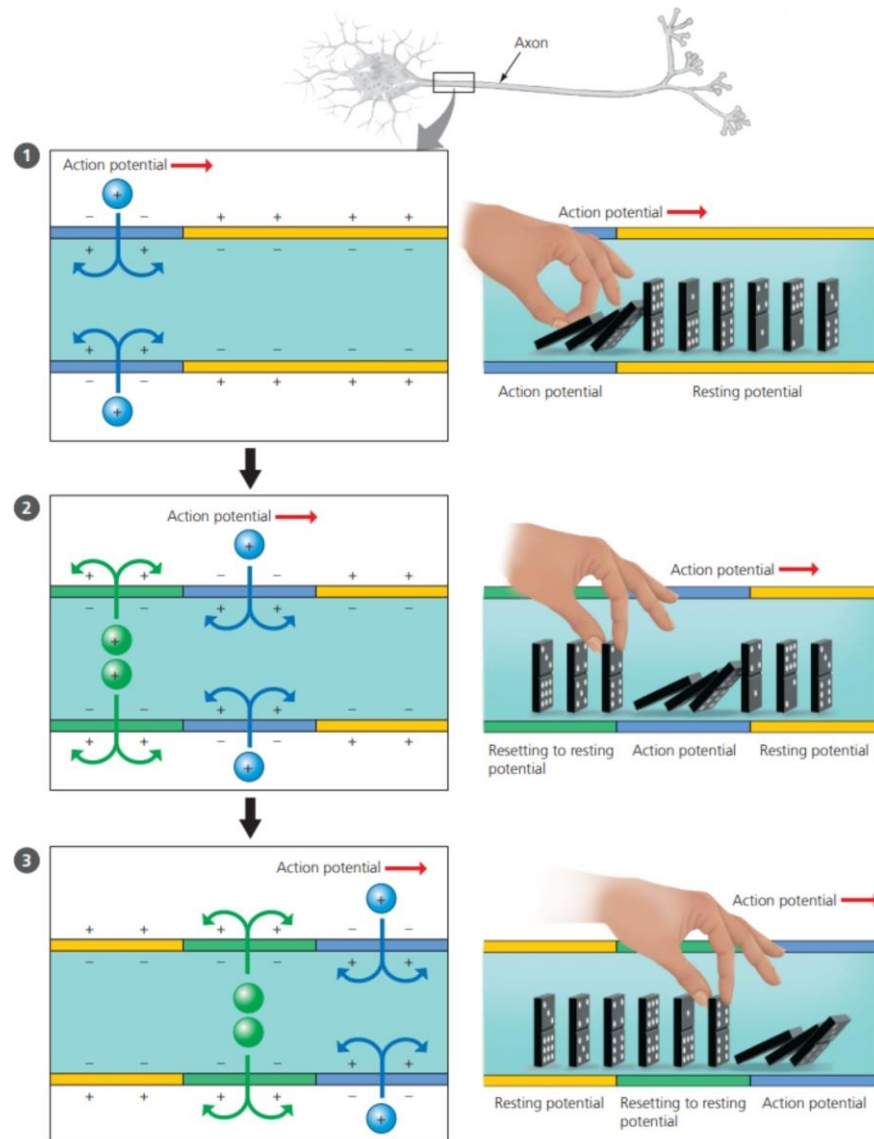
Generation of an action potential



Propagation of the Signal

- An action potential is a change from the resting potential at a specific place along a neuron.
 - A nerve signal starts as an action potential generated near a neuron's cell body.
 - To function as a long-distance signal, this local change must be propagated, that is, passed along the length of the neuron. Nerve signals are self-propagating, meaning that **a voltage change in one location on its axon causes a voltage change in the next location on its axon**.
 - **Figure 27.4** shows the changes that occur as an action potential moves down a neuron's axon.

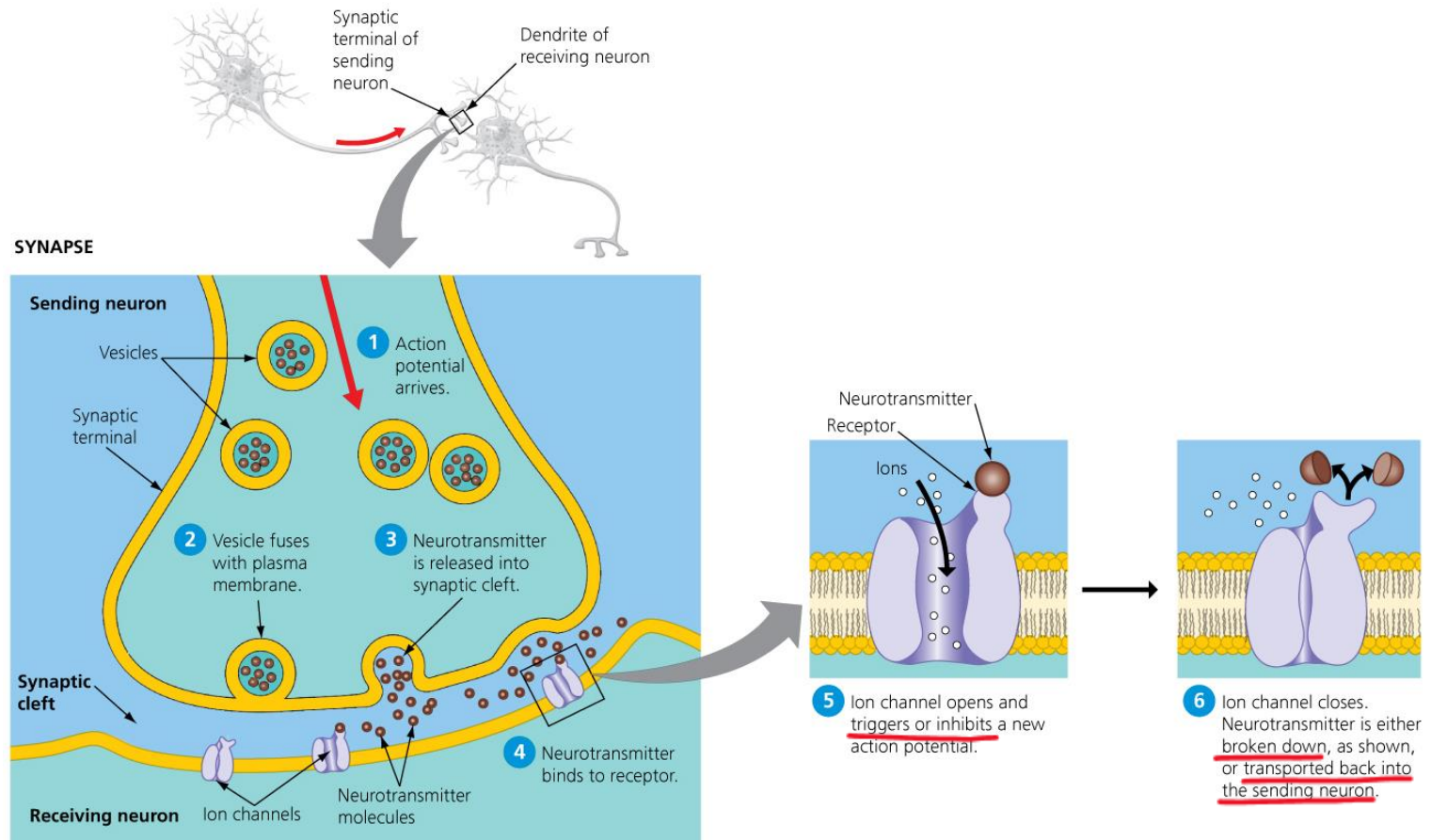
Propagation of an action potential along an axon



Passing a Signal from a Neuron to a Receiving Cell

- A **synapse** is a relay point between a neuron and a receiving cell. The receiving cell can be another neuron or another cell type such as a muscle cell or a hormone-secreting cell. Synapses come in two varieties.
 1. In an **electrical synapse**, electric currents pass directly from one neuron to the next and provide very rapid responses.
 2. **Chemical synapses** provide more diverse and subtle responses than electrical synapses.
 - **Figure 27.5** shows the events that occur at a synapse between two neurons.

Neuron communication at a synaptic cleft



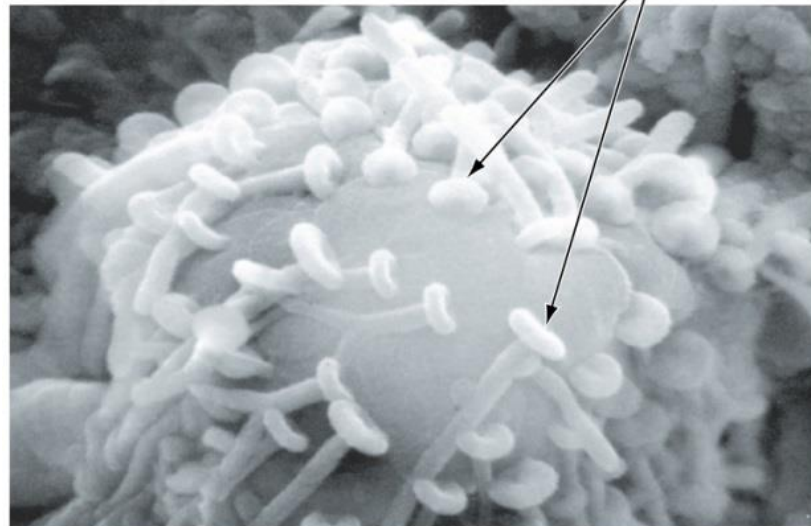
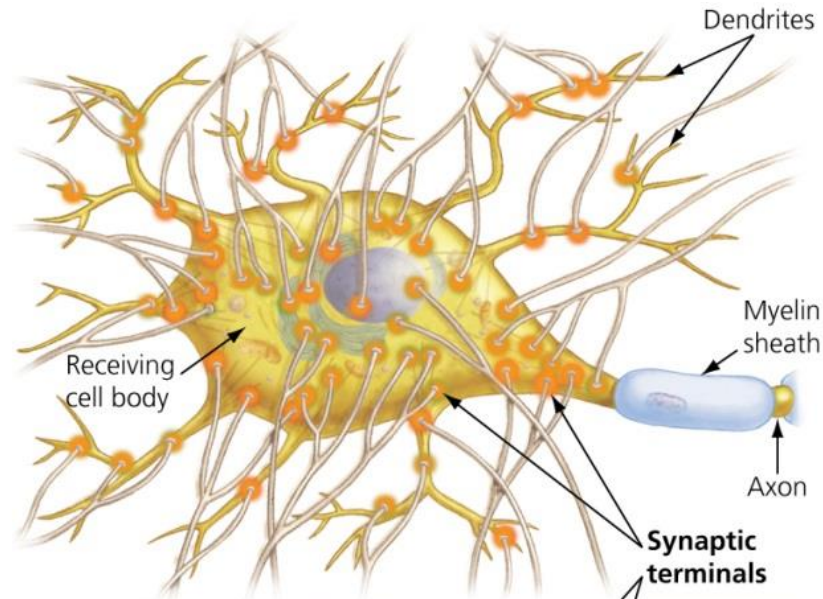
Chemical Synapses (1 of 2)

- Neurotransmitters result in many different actions, and a single neurotransmitter, such as dopamine, can play a role in pleasure, attention, memory, sleep, movement, and learning.
 - Some types of depression have been linked to reduced levels of neurotransmitters, specifically norepinephrine and serotonin.
 - The degenerative illness Parkinson's disease is associated with a lack of the neurotransmitter dopamine, while an excess of dopamine is linked to schizophrenia.

Chemical Synapses (2 of 2)

- A neuron may receive input from hundreds of other neurons via thousands of synaptic terminals.
 - The inputs can be highly varied because each sending neuron may secrete a different quantity or kind of neurotransmitter.
 - The sum total of excitatory and inhibitory inputs determines whether or not the neuron fires.
 - These factors account for the nervous system's ability to process huge amounts of complex stimuli and formulate appropriate responses.

A neuron's multiple synaptic inputs



SEM 5,000x

Drugs and the Brain (1 of 3)

- Many common substances, such as caffeine, nicotine, and alcohol affect the actions of neurotransmitters.
 - Caffeine counters the effects of neurotransmitters that normally suppress nerve signals.
 - Prescription drugs used to treat psychological disorders often alter the effects of neurotransmitters.
 - A drug may physically **block a receptor**, preventing a neurotransmitter from binding, reducing its effect.
 - Other drugs **block the reuptake of neurotransmitters** from the sending neuron, effectively increasing the time it is active at the synapse.

Drugs and the Brain (2 of 3)

- Stimulants such as amphetamines and cocaine increase the release and availability of norepinephrine and dopamine at synapses. Abuse of these drugs can therefore produce symptoms resembling schizophrenia.
- LSD and mescaline may produce their hallucinatory effects by activating serotonin and dopamine receptors.
- The active ingredient in marijuana binds to brain receptors normally used by other neurotransmitters that seem to play a role in pain, depression, appetite, memory, and fertility.

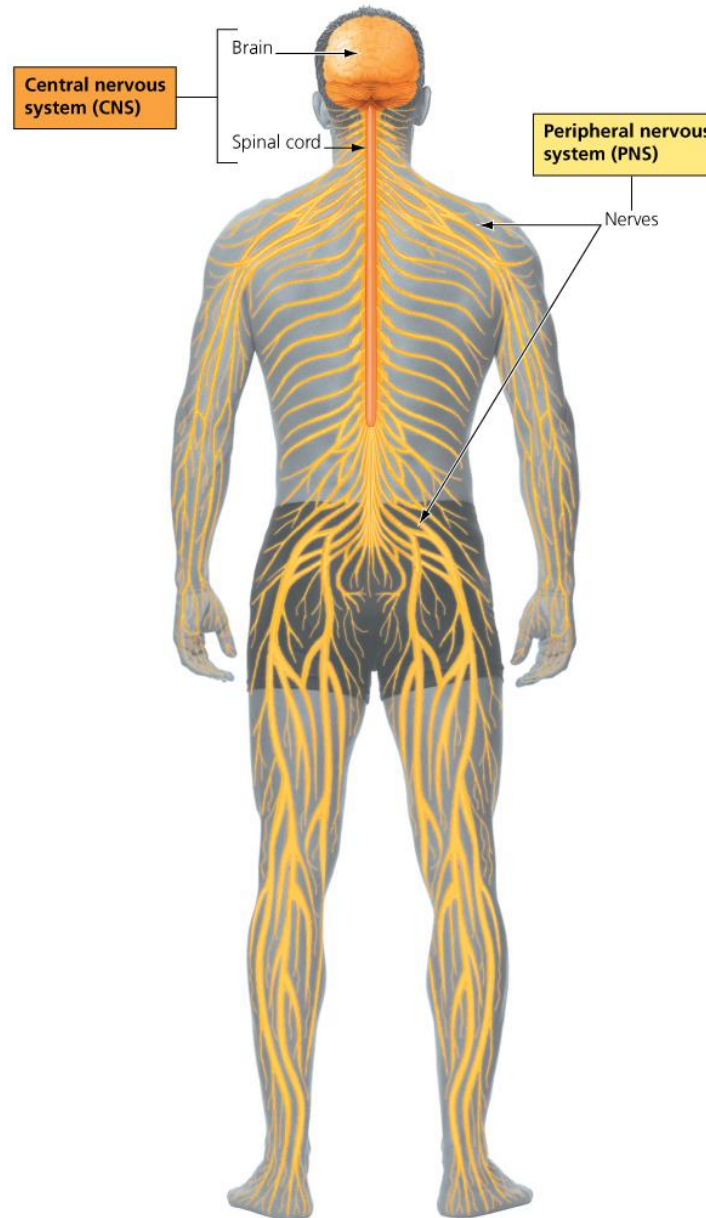
Drugs and the Brain (3 of 3)

- **Opiates**— morphine, codeine, and heroin—bind to the receptors for the neurotransmitter endorphin, reducing pain and producing euphoria. Thus, opiates are usually prescribed to relieve pain.
- While the drugs mentioned can increase alertness and a sense of well-being or reduce physical and emotional pain, they also have the potential to disrupt the brain's finely tuned neural pathways, altering the chemical balances that are the product of millions of years of evolution.

The Human Nervous System: A Closer Look

- Although there is remarkable uniformity throughout the animal kingdom in the way nerve cells function, there is great variety in how nervous systems as a whole are organized.
 - Vertebrate nervous systems are diverse in structure and level of sophistication.
 - However, some features are common to the nervous systems of all vertebrates. All have a nervous system that is concentrated at the head end, and all have a central nervous system (CNS) distinct from a peripheral nervous system (PNS).

A vertebrate nervous system (back view)



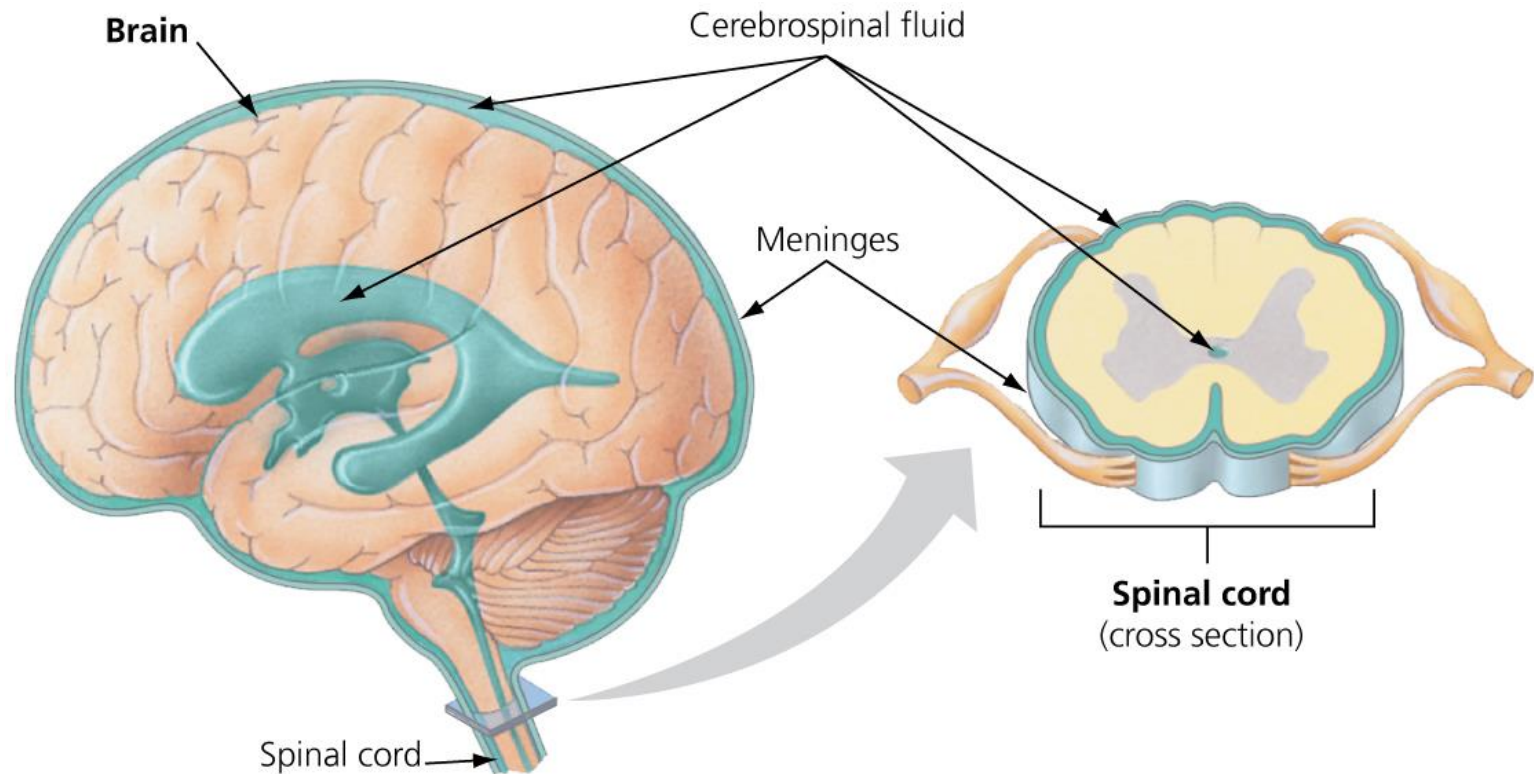
The Central Nervous System (1 of 2)

- The central nervous system (CNS) interprets signals coming from peripheral nerves and sends out responses through other peripheral nerves.
 - The CNS consists of **two parts** in vertebrates.
 1. The **brain** functions as the master control center for the nervous system, integrating data from the sense organs and sending out commands from motor control centers.
 2. The **spinal cord** is a bundle of nerve fibers running through the spinal column, forming a communication conduit between the brain and the rest of the body.

The Central Nervous System (2 of 2)

- Injury to the CNS is often permanent, because the neurons of the CNS cannot be repaired. Trauma to the brain can have a variety of effects—such as personality changes, coma, or death—depending on what part of the brain is affected.
- The brain and spinal cord are surrounded by and contain spaces that are filled with **cerebrospinal fluid**, a liquid that is formed in the brain by filtration of arterial blood. This liquid supplies nutrients and hormones to the CNS and carries away wastes.
- Protecting the brain and spinal cord are layers of connective tissue called **meninges**.

Fluid-filled spaces of the vertebrate CNS



The Peripheral Nervous System (1 of 2)

- The **output** components of the vertebrate PNS are divided into two functional units.
 1. Neurons of the **motor system** carry signals to skeletal muscles, mainly in response to external stimuli.
 2. The **autonomic nervous system** regulates the internal environment by controlling smooth and cardiac muscles and the glands of several body systems. This control is generally involuntary.

Functional divisions of the vertebrate PNS

PERIPHERAL NERVOUS SYSTEM		
Motor system (voluntary)	Autonomic nervous system (involuntary)	
	Parasympathetic division 	Sympathetic division 
Control of skeletal muscle	"Rest and digest"	"Fight or flight"

The Peripheral Nervous System (2 of 2)

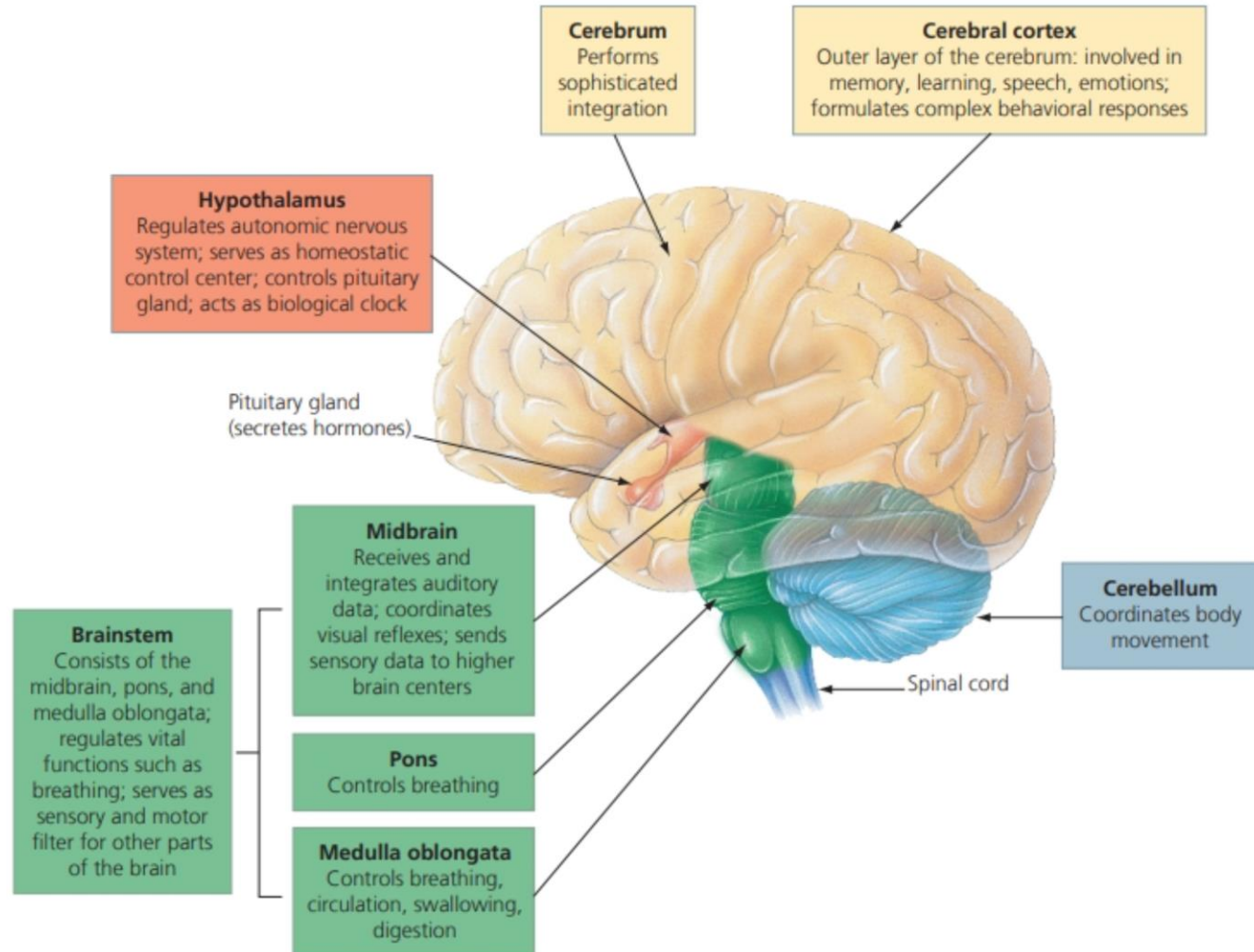
- The autonomic nervous system contains two sets of neurons with opposing effects on most organs.
 1. The **parasympathetic division** primes the body for activities related to long-term needs of the body, including stimulating the digestive organs, maintaining an active immune response, decreasing the heart rate, and narrowing the bronchi, related to a decreased breathing rate.
 2. The **sympathetic division** tends to have the opposite effect, preparing the body for intense, energy-consuming activities, such as fighting or fleeing.

The Human Brain

- The human brain is more powerful than the most sophisticated computer. It consists of up to 100 billion intricately organized neurons and a much larger number of supporting cells.
- **Figure 27.10** highlights some major parts of the brain and associated functions.

Some major parts of the human brain

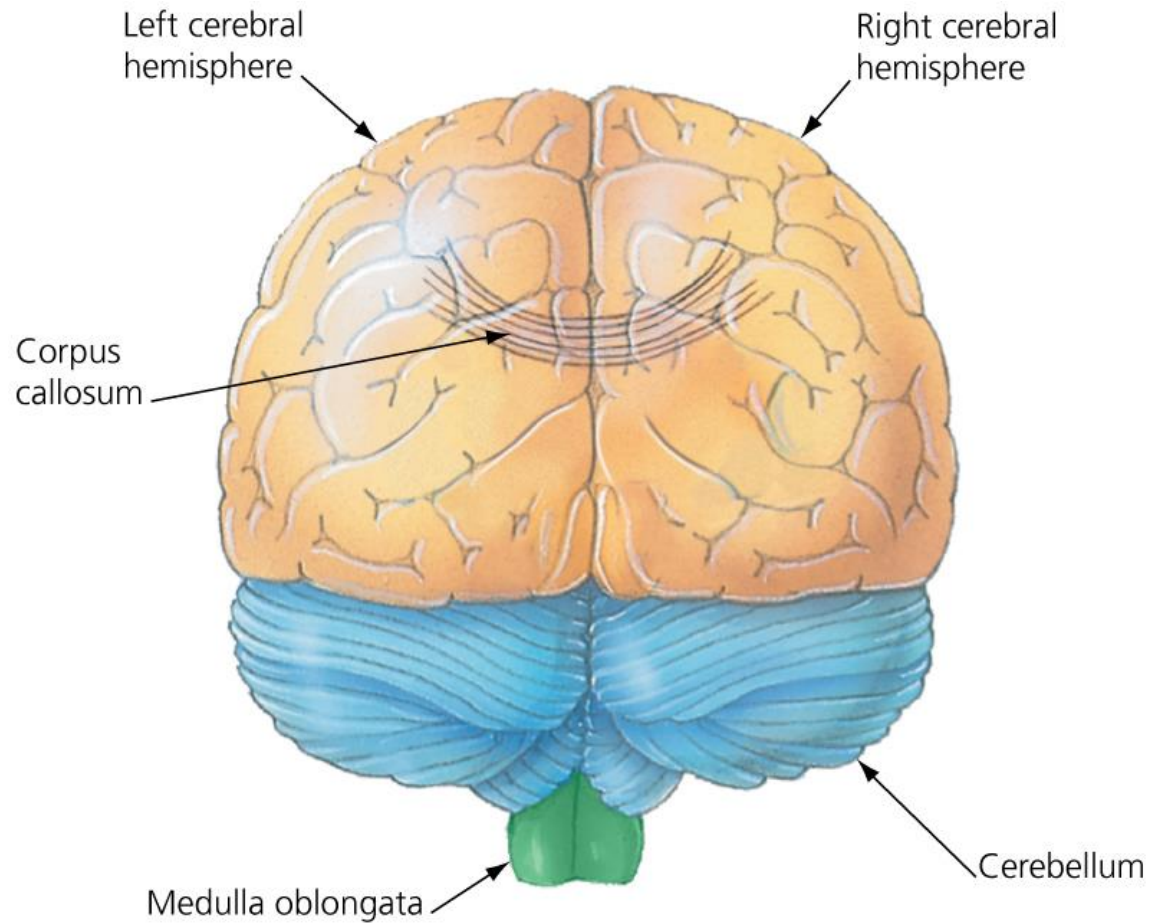
▼ **Figure 27.10** Some major parts of the human brain.



The Cerebral Cortex (1 of 4)

- The cerebral cortex is divided into right and left hemispheres.
 - Each half receives information from and controls the movement of the opposite side of the body.
 - The **corpus callosum**, a thick cable of nerve fibers, bridges the two sides and enables them to process information together. The corpus callosum is often reduced in size and sometimes even absent in autistic individuals.

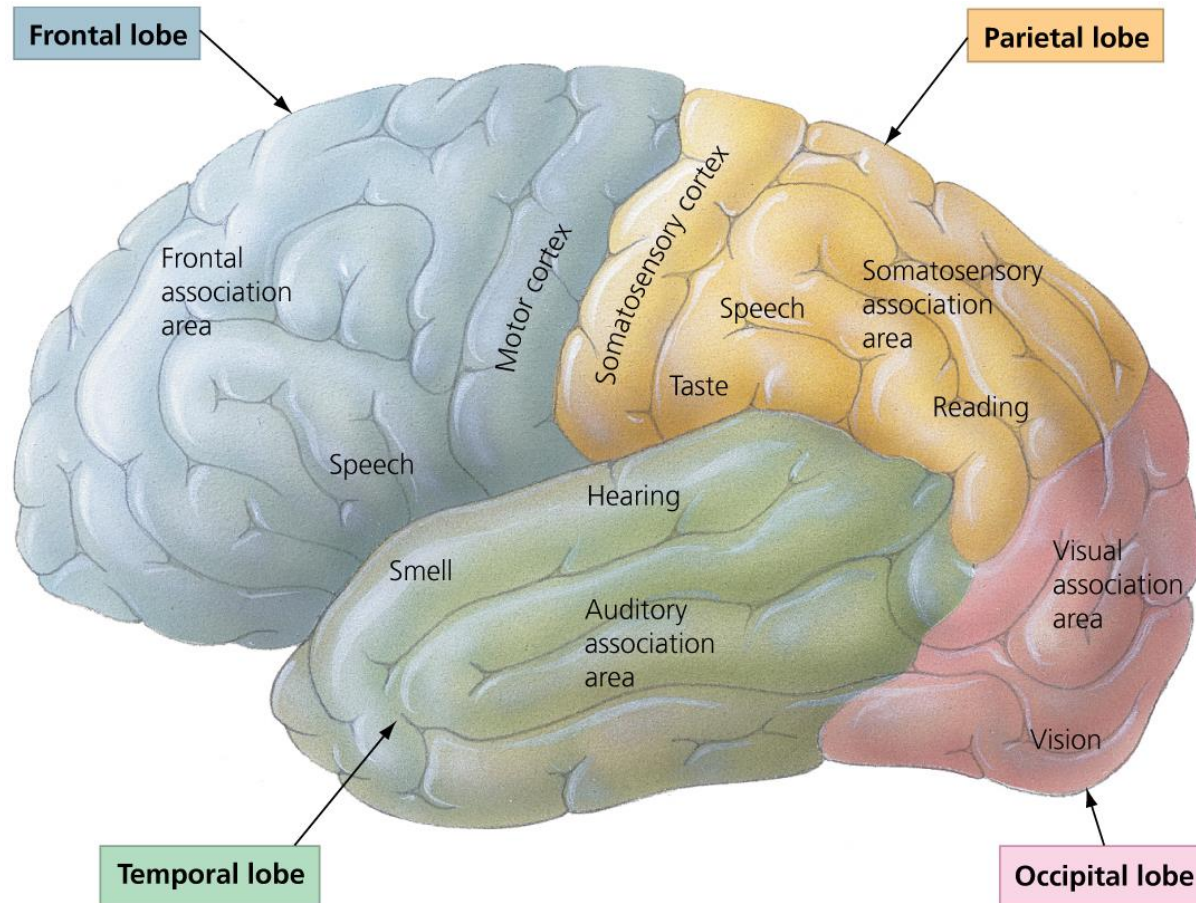
A rear view of the brain



The Cerebral Cortex (2 of 4)

- Both the right and left hemispheres of the cerebral cortex have four major lobes, each named for a nearby skull bone: the frontal, parietal, temporal, and occipital lobes.
- These four lobes are represented by different colors in **Figure 27.12**, where their specific functions are noted.

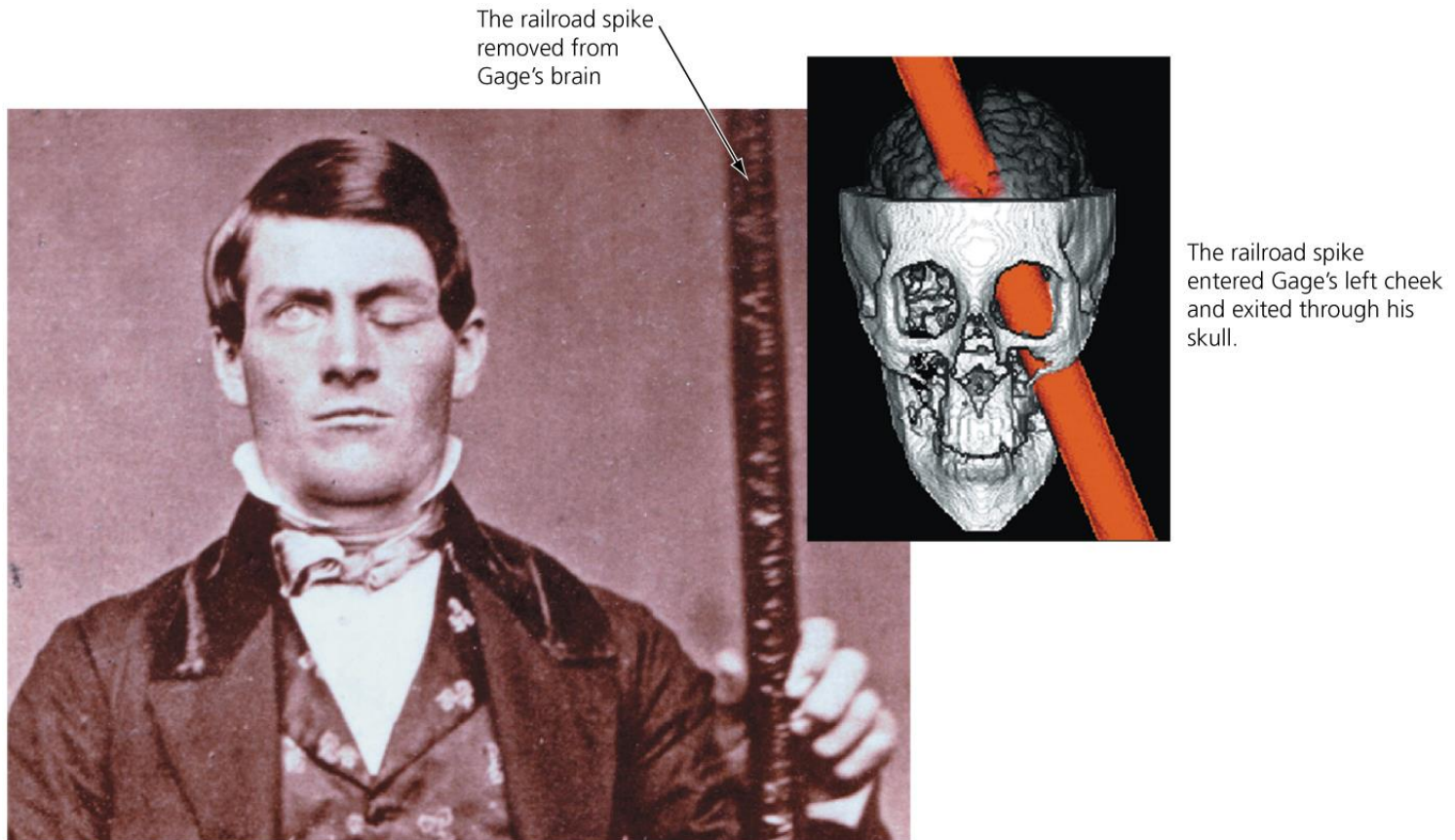
Functional areas of the cerebrum's left hemisphere



The Cerebral Cortex (3 of 4)

- Injuries to the brain provide scientists with clues about the function of certain association areas. A well-known example is **Phineas Gage**.
 - In 1848, an explosion propelled a 3-foot-long railroad spike through his head.
 - Incredibly, after recovery from the trauma, it appeared he was able to **think normally**. However, **changes in his personality** soon appeared, including a propensity for **meanness and impulsiveness**.
 - People with frontal lobe injuries often exhibit **irrational decision making and difficulty processing emotions**, similar to Phineas Gage after his injury.

Phineas Gage's accident



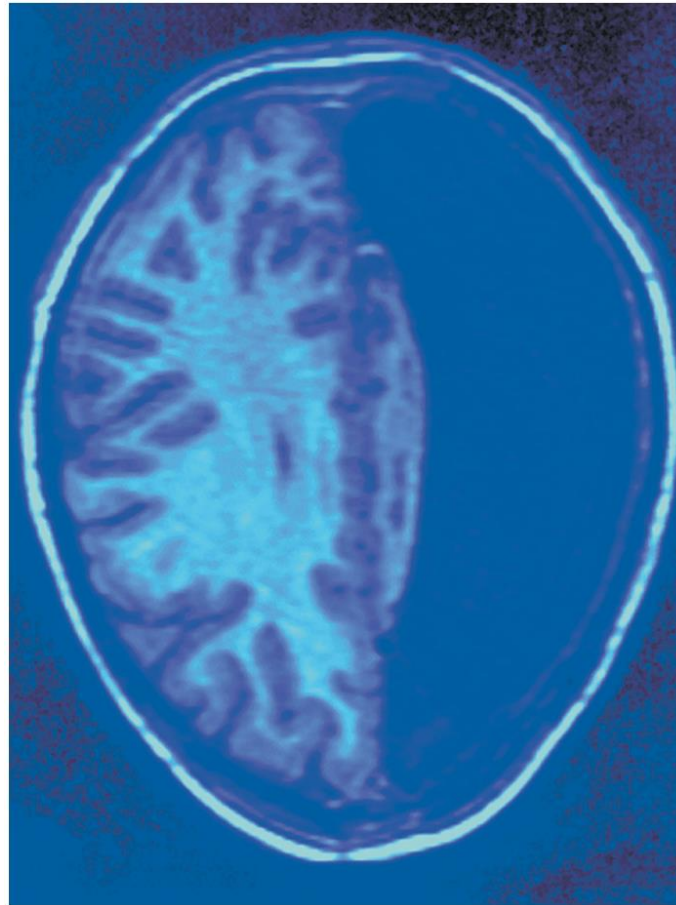
The Cerebral Cortex (4 of 4)

- In a typical brain, the association areas of the frontal lobe are some of the last areas to mature.
 - One association area, involved in decision-making, likely continues to develop into our 20s. This might account for riskier decisions made by teens and young adults compared to older adults.
 - A delay in the maturation of the brain could explain a common brain disorder. **Attention deficit hyperactivity disorder (ADHD)** is characterized by a lack of focus, impulsive behavior, and restlessness. Studies of the brains of youths with ADHD found maturation of the brain is delayed by about 3 years.

Neuroplasticity

- Evidence from brain surgery patients indicates that our brain's wiring is not necessarily fixed.
 - One of the most radical surgical alterations of the brain is **hemispherectomy**, the removal of almost one-half of the brain. Incredibly, with just half a brain, hemispherectomy patients recover quickly, often leaving the hospital within a few weeks. However, the side of the body opposite the surgery always has **partial paralysis**.
 - Recovery after hemispherectomy is a striking example of the brain's remarkable **neuroplasticity**, its ability to **reorganize its neural connections**.

Hemispherectomy



Neurological Disorders (1 of 3)

- Neurological disorders, diseases of the nervous system, often involve the brain.
 - **Alzheimer's disease** is a form of mental deterioration, or dementia, characterized by confusion, memory loss, and a variety of other symptoms.
 - One particularly common characteristic of Alzheimer's disease is the **loss of neurons in the temporal lobe**.
 - Its incidence is usually age related, rising from about 12% at age 65 to almost 50% at age 85.

Neurological Disorders (2 of 3)

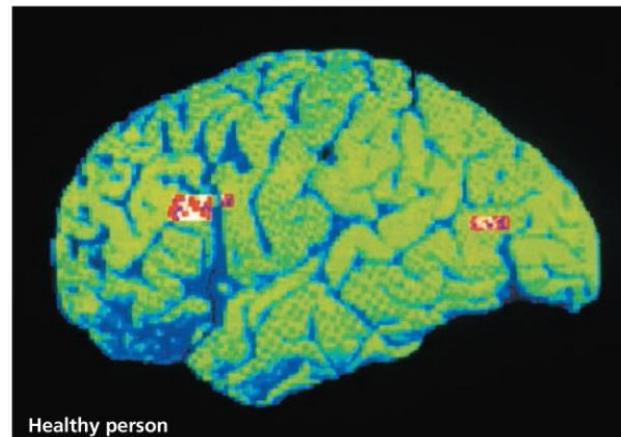
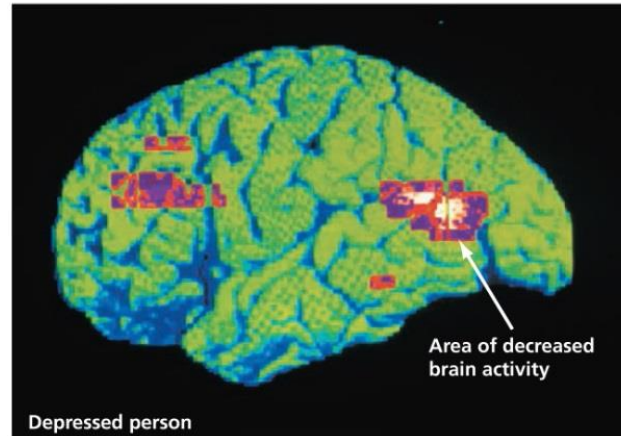
- In **chronic traumatic encephalopathy** (CTE) the frontal lobe and temporal lobes frequently show significant deterioration.
 - CTE is caused by repeated brain trauma, particularly concussions. Numerous deceased football players have been diagnosed with CTE posthumously.
 - Like Alzheimer's disease, early symptoms include personality changes. Depression, loss of impulse control, and eventually, memory loss and dementia are prominent features of CTE.

Neurological Disorders (3 of 3)

- Other neurological disorders alter the activity of certain brain regions. Two broad forms of depressive illness are known.
 1. People with **major depression** may experience sadness, loss of interest in pleasurable activities, changes in body weight and sleep patterns, and suicidal thoughts. Major depression is extreme and persistent, leaving the sufferer unable to live a normal life.
 2. **Bipolar disorder**, or **manic-depressive disorder**, involves extreme mood swings.

Brain activity in a depressed person and a healthy person

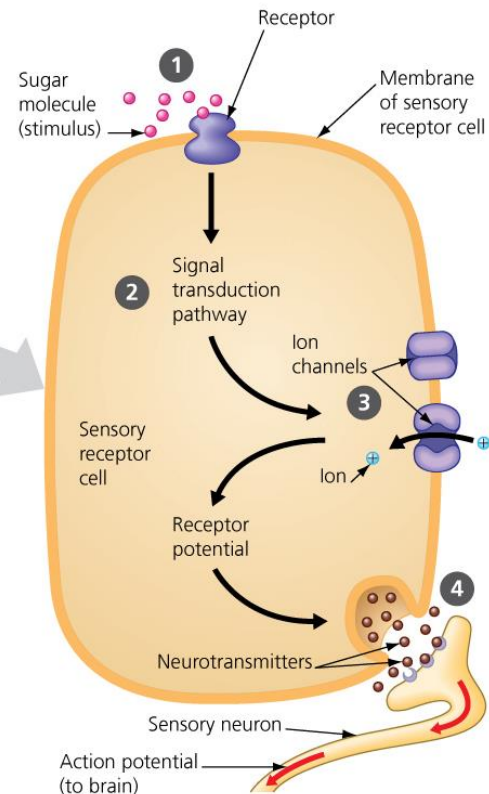
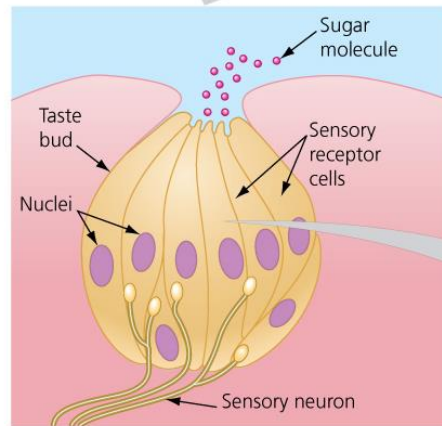
certain areas of the brain.



The Senses: Sensory Input

- **Sensory structures** gather information and pass it on to the central nervous system (CNS).
 - Sensory receptors detect stimuli such as light, sounds, cold, heat, and touch and send information to the CNS by triggering action potentials.
 - In stimulus detection, the receptor cell converts one type of signal (the stimulus) to an electrical signal.
 - This conversion of the signal, called **sensory transduction**, occurs as a change in the membrane potential of the receptor cell.
 - **Figure 27.16** shows sensory receptors in a taste bud detecting sugar molecules.

Converting a chemical stimulus to an electrical signal in a human taste bud



Converting a Stimulus to an Electrical Signal

- A change in membrane potential is called the **receptor potential**. In contrast to **action potentials**, which are all-or-none phenomena, **receptor potentials vary in intensity**; the stronger the stimulus, the stronger the receptor potential.
- **Sensory adaptation** is the tendency of some sensory receptors to become **less sensitive** when they are stimulated repeatedly. This keeps the body from continuously reacting to normal background stimuli.

Types of Sensory Receptors (1 of 2)

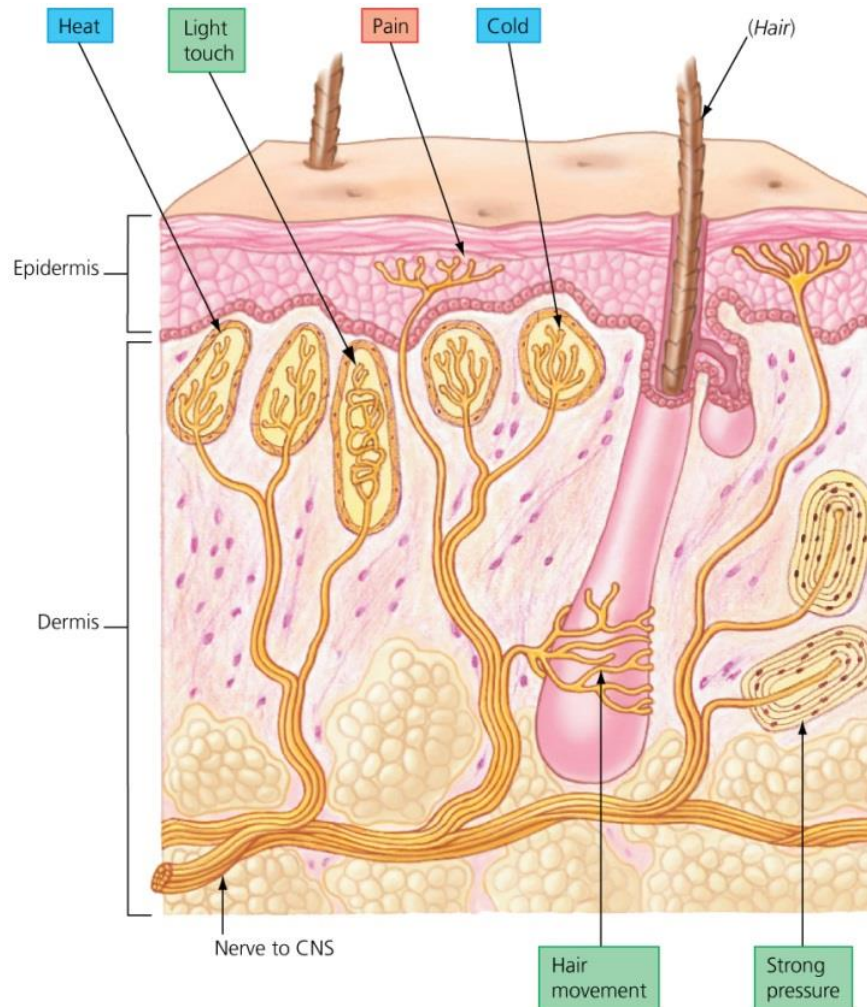
- Sensory receptors can be grouped into five categories, which work in various combinations to produce the five human senses.
 1. **Pain receptors** may respond to excessive heat or pressure or to chemicals released from damaged or inflamed tissues.
 2. **Thermoreceptors** detect heat or cold.
 3. **Mechanoreceptors** are stimulated by various forms of mechanical energy such as touch and pressure, stretching, motion, and sound.

Types of Sensory Receptors (2 of 2)

4. **Chemoreceptors** include the sensory cells in your nose and taste buds, which respond to chemicals in the external environment or body fluids.
5. **Electromagnetic receptors** are sensitive to energy of various wavelengths and respond to magnetism or light (**photoreceptors**).

Figure 27.17 shows a section of human skin to reveal why the surface of our body is sensitive to such a variety of stimuli.

Sensory receptors in the human skin

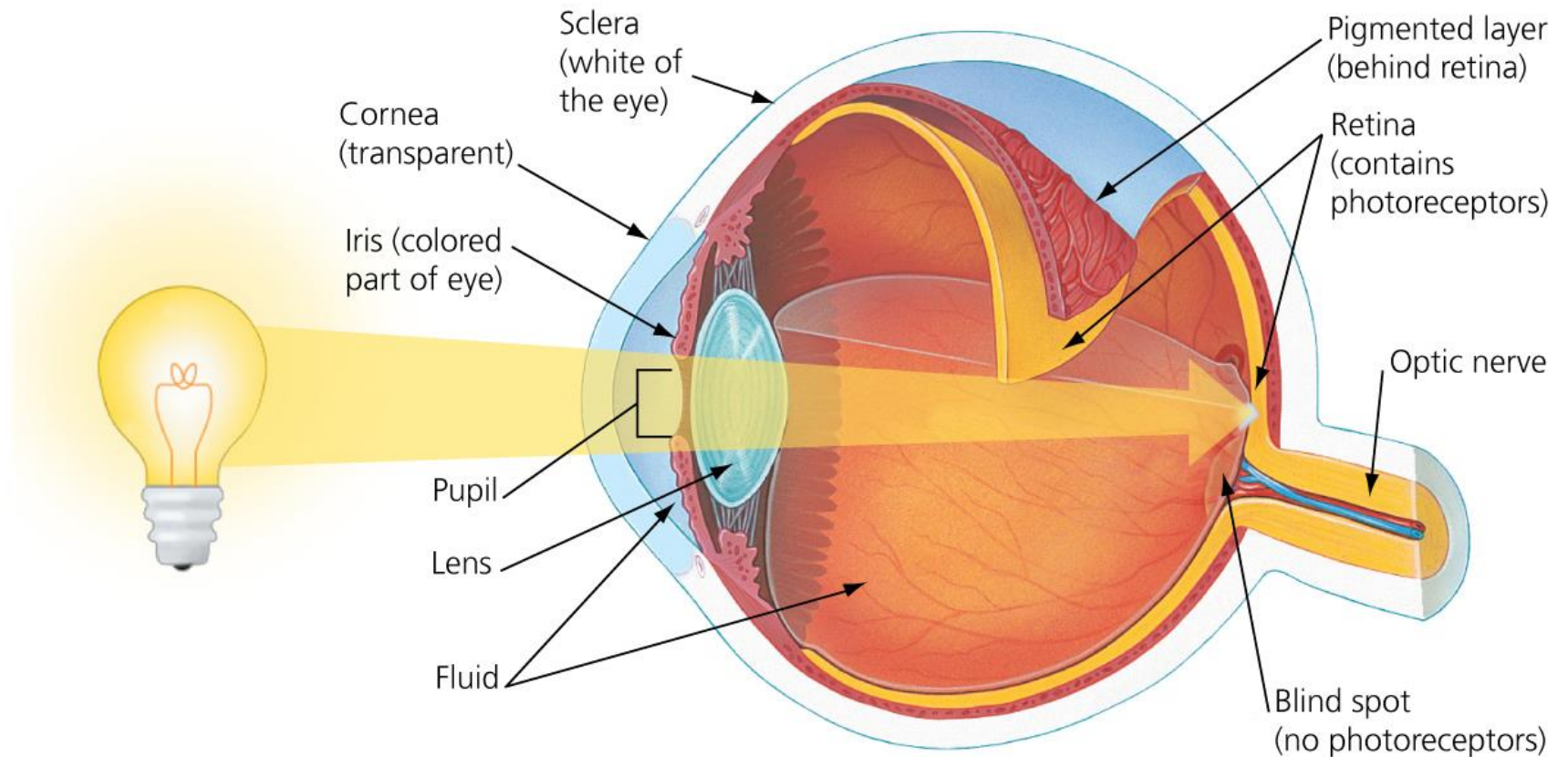


Vision: Lighting the Retina

- Human eyes are able to detect a multitude of colors, form images of faraway objects, and respond to minute amounts of light energy.
 - The human eye consists of
 - a tough outer covering, the whitish sclera,
 - a transparent **cornea** in front of the disk-like **lens**,
 - a pigmented **iris** with a circular opening, the **pupil**, and
 - the **retina**, at the back of the eyeball, where photoreceptors respond to light.

Sclera: 공막

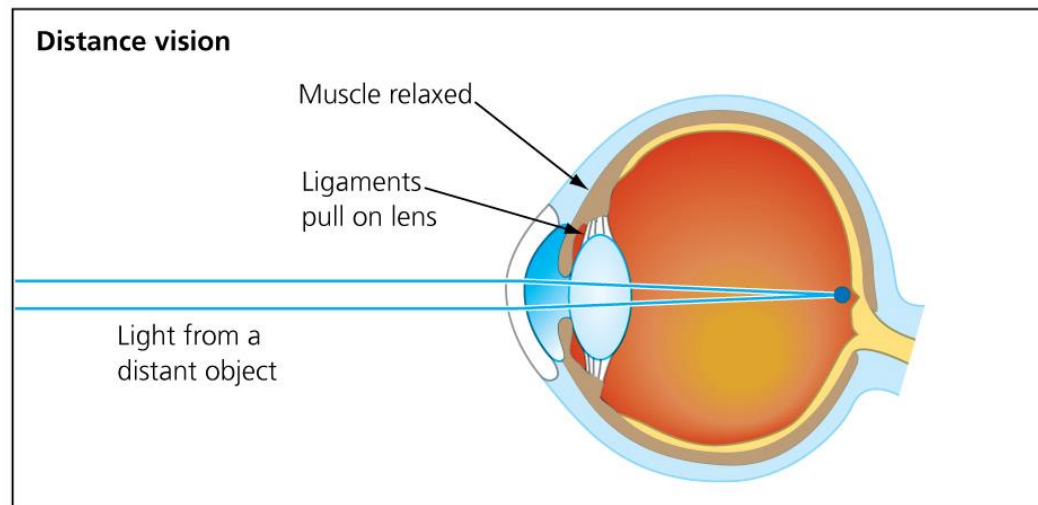
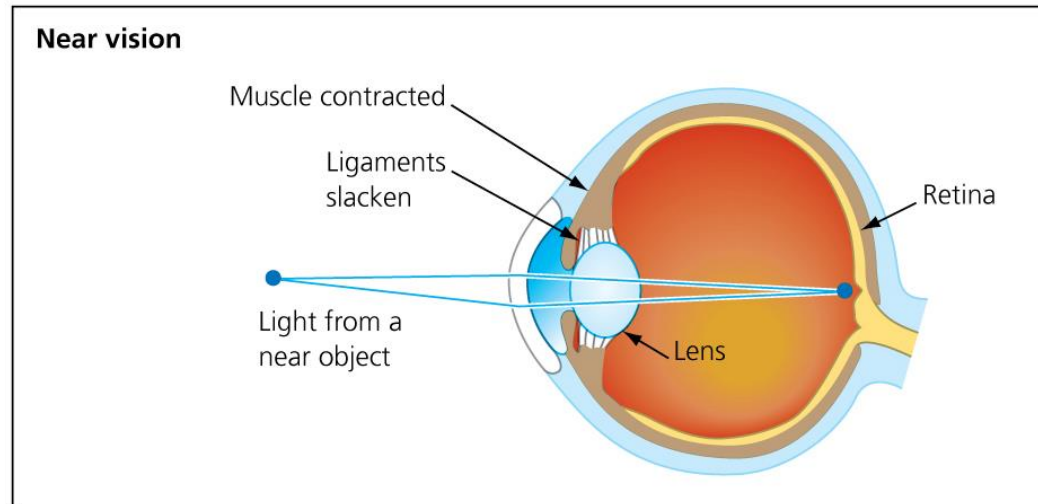
The path of light to the retina



Lighting the Retina

- The cornea lets light into the eye and helps focus light.
- The iris regulates the size of the pupil to adjust to varying light.
- The lens of the eye changes shape and focuses light onto the retina by bending light rays.

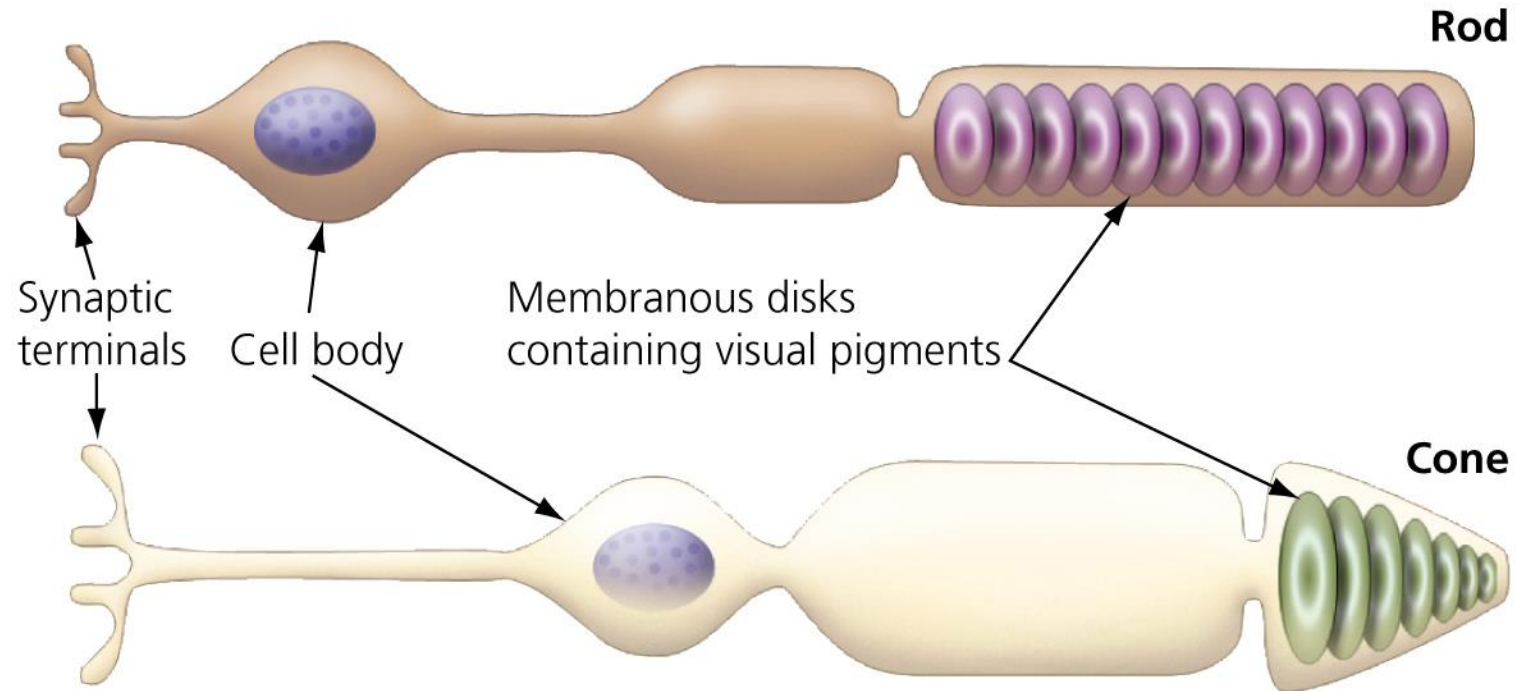
How the lens of the eye focuses light



Photoreceptors (1 of 2)

- The human retina contains two types of photoreceptors named for their shapes.
 1. **Rods** are extremely sensitive to light and enable us to see in dim light, but only in shades of gray.
 2. **Cones** are stimulated by bright light and can distinguish color but contribute little to night vision. There are **three types of cones**, each containing a different type of photoreceptor protein, which absorbs blue, green, or red light.

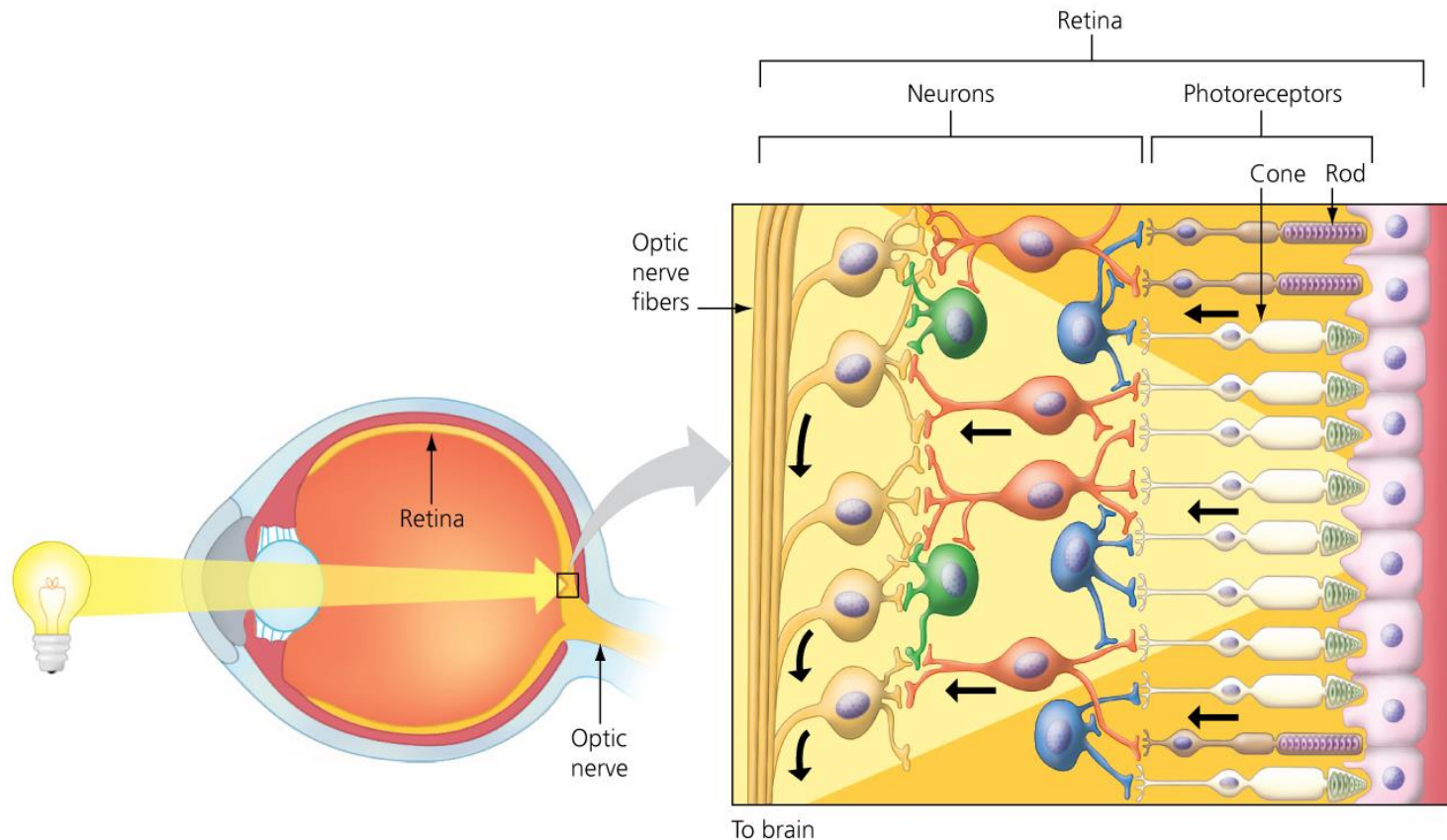
Photoreceptor cells



Photoreceptors (2 of 2)

- Like all receptor cells, rods and cones convert a stimulus to an electrical signal.
- In **Figure 27.21**, we see light reaching these photoreceptors on the right side of the illustration.
 - As they absorb light, visual pigments change chemically, and the change alters the permeability of the cell's membrane.
 - Other retinal neurons start to integrate these receptor potentials and generate action potentials that travel along the **optic nerve** to the brain.

The pathway that light travels to activate photoreceptors and stimulate the optic nerve

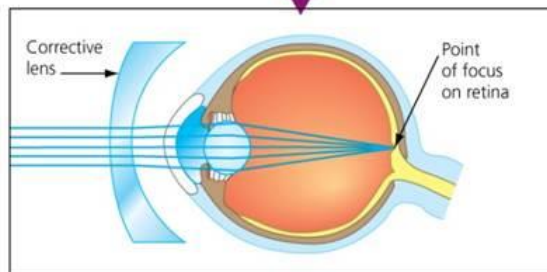
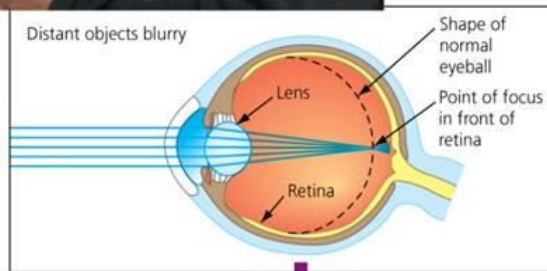


Imperfect Vision (2 of 2)

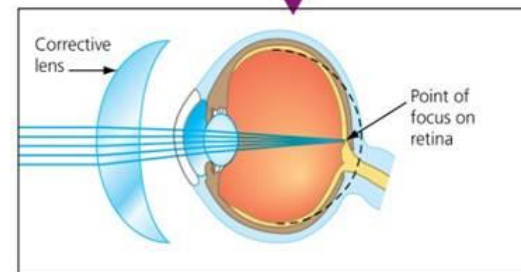
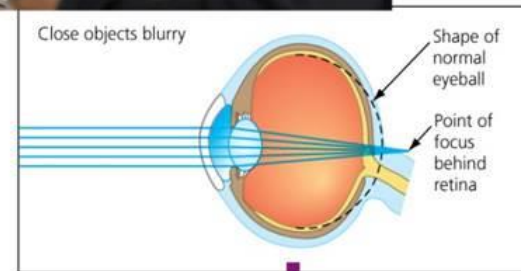
- The most common visual problems are
 - **nearsightedness**, the inability to focus well on distant objects,
 - **farsightedness**, the inability to focus well on close objects, and
 - **astigmatism**, blurred vision caused by a misshapen lens or cornea.
 - Surgical procedures are an option for treating vision disorders. In a laser-assisted surgery known as LASIK, the cornea is reshaped to change its focusing ability.

Astigmatism: 난시
Misshapen: 기형의

A nearsighted eye and a farsighted eye



(a) A nearsighted eye (eyeball too long).
Nearsightedness is corrected by lenses that are thinner toward the middle.



(b) A farsighted eye (eyeball too short).
Farsightedness is corrected by lenses that are thicker in the middle.

Hearing (1 of 4)

- **Figure 27.24** shows how sound travels through the ear to where the receptors are located.
 1. The bendable pinna acts like a funnel, focusing sound waves into the auditory canal. The pinna and auditory canal are considered the **outer ear**.
 2. The sound waves then reach a sheet of tissue called the **eardrum**, which separates the outer ear from the middle ear. The sound waves cause the eardrum to vibrate.
 3. From the eardrum, the vibrations are concentrated as they pass through the hammer, anvil, and stirrup—three bones of the middle ear.

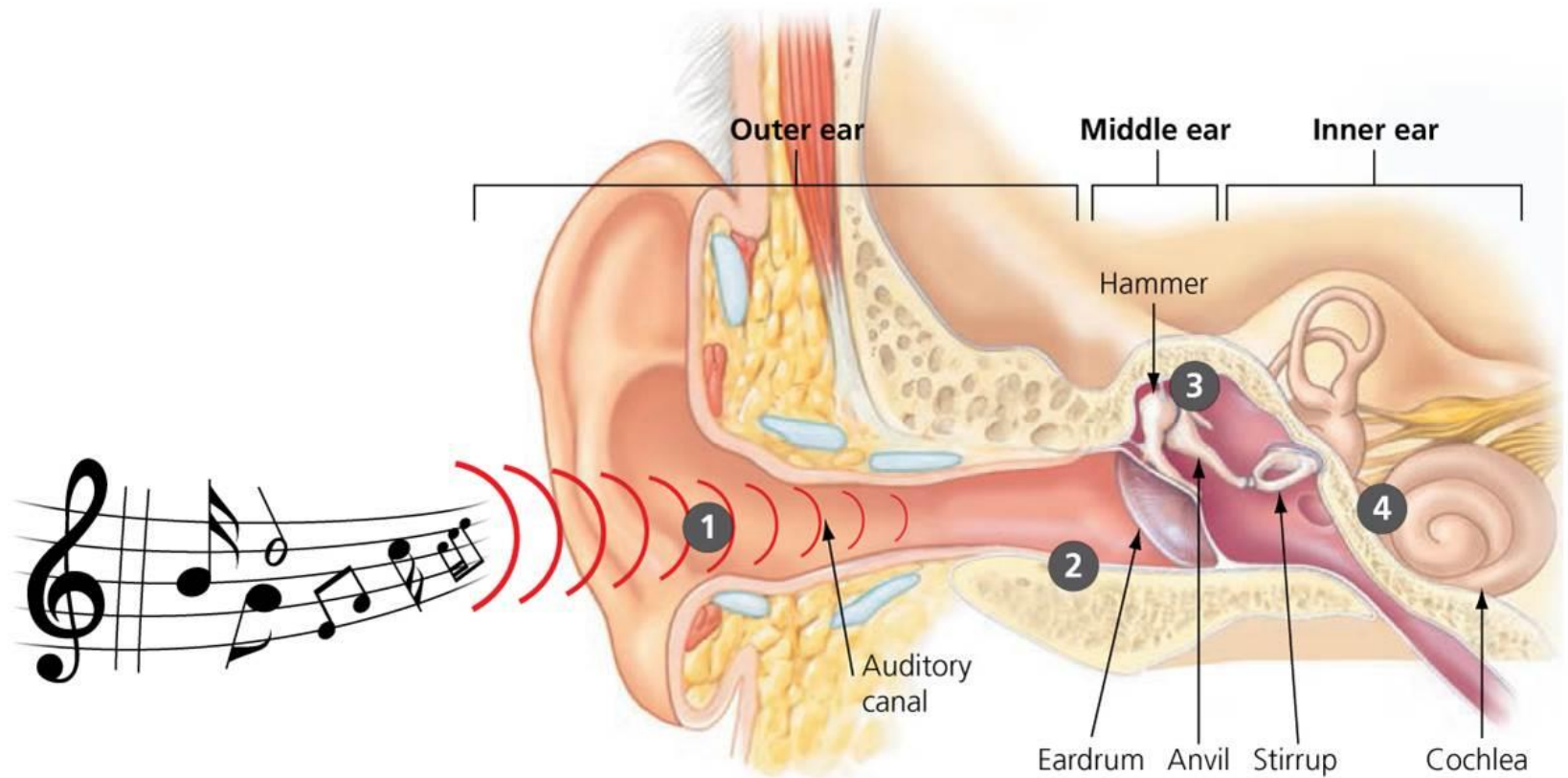
Pinna: 귓바퀴

Auditory canal: 이도

Eardrum: 고막

Stirrup: 등자

How sound travels to the cochlea



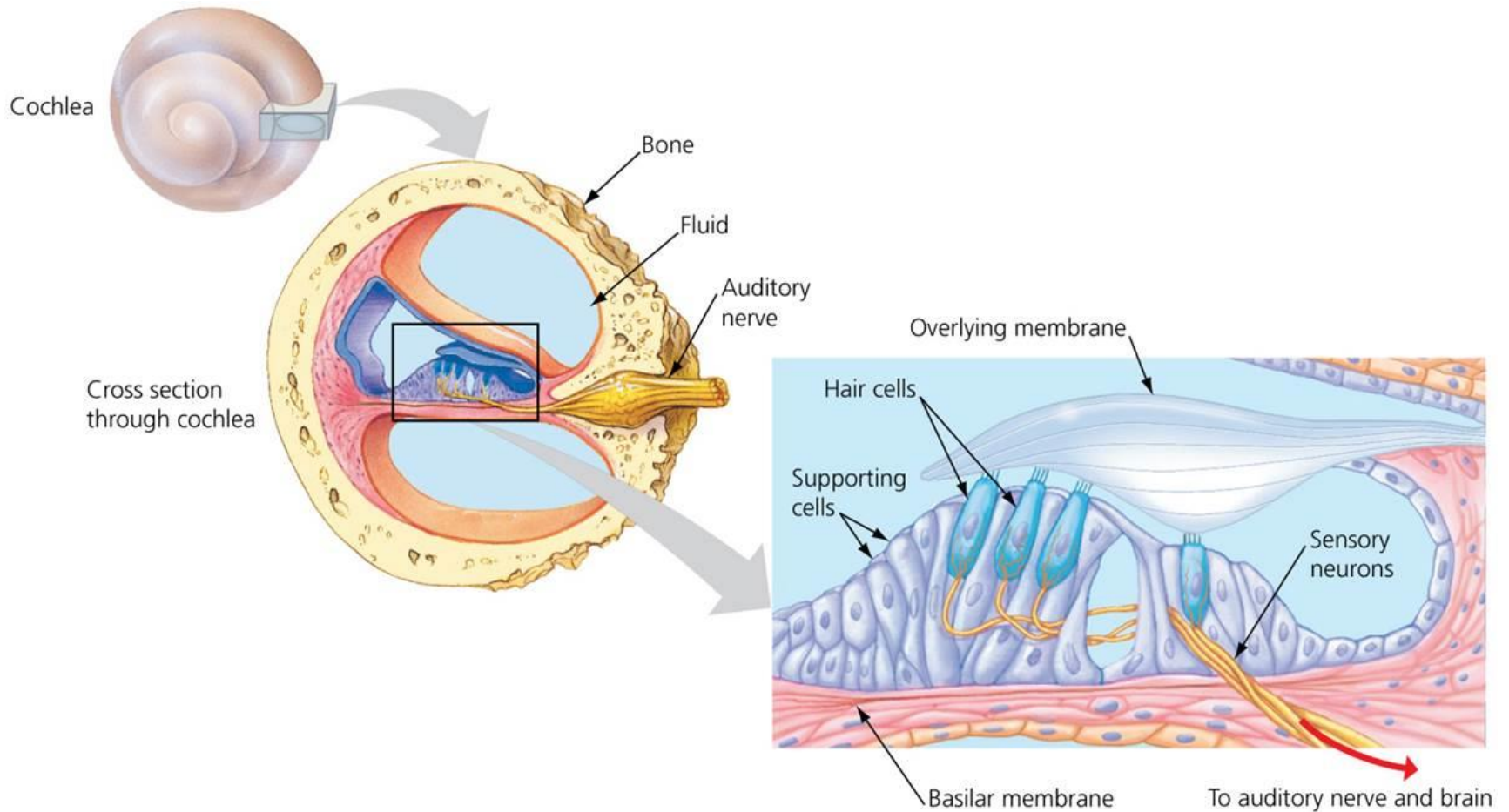
Hearing (2 of 4)

4. The stirrup transmits the vibrations to a sheet of tissue between the middle ear and the third part of the ear, the inner ear.
- As the bones vibrate, they push on the membrane to generate vibrations in the fluid that fills the inner ear.
 - A component of the inner ear is the cochlea, a coiled, fluid-filled structure containing our hearing organ, where sensory receptors are located and sound is detected. In the cochlea, receptor cells known as hair cells are anchored on the basilar membrane. The vibrations traveling from the middle ear produce pressure waves within the cochlea that cause the basilar membrane to vibrate.

Cochlea: 달팽이관

Basilar membrane: 기저막

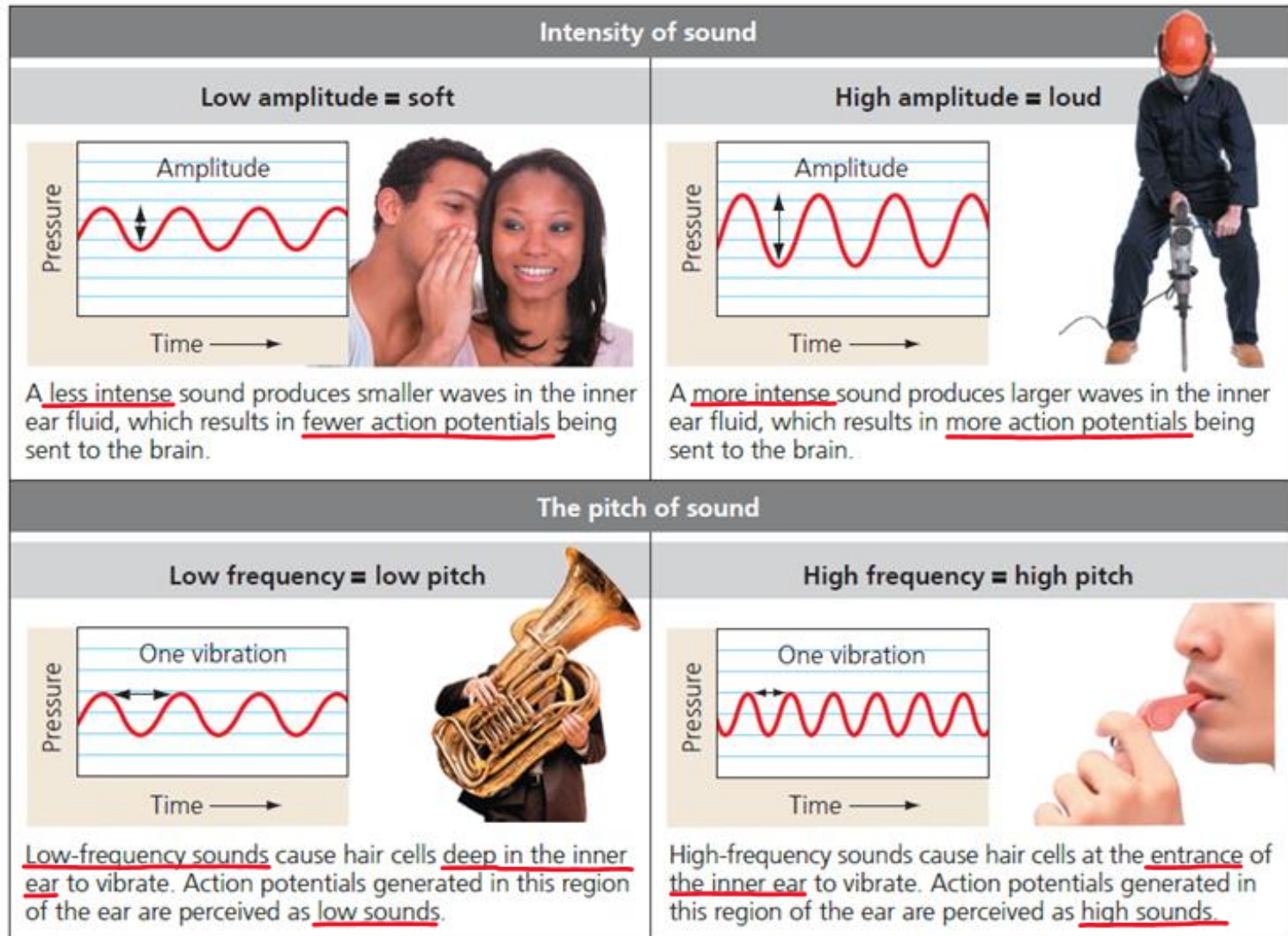
The cochlea



Hearing (3 of 4)

- As the basilar membrane vibrates, hair-like projections on the hair cells bend against an overlying membrane.
- The hair cell develops a receptor potential and releases neurotransmitters at its synapse with a sensory neuron, which in turn sends action potentials through the auditory nerve to the brain.
- The amplitude of sound waves affects how vigorously the hair cells vibrate.
- You can distinguish particular tones (pitches) because various frequencies of sound waves cause different hair cells to bend and vibrate.

The characteristics of sound and perceived volume and pitch



Hearing (4 of 4)

- Deafness, the inability to hear, can be present at birth or acquired after
 - middle ear infections,
 - injury, such as a ruptured eardrum,
 - stiffening of the middle-ear bones (a common age-related problem), or
 - overexposure to loud noises.
- One treatment for deafness or major hearing loss is a device called a cochlear implant.

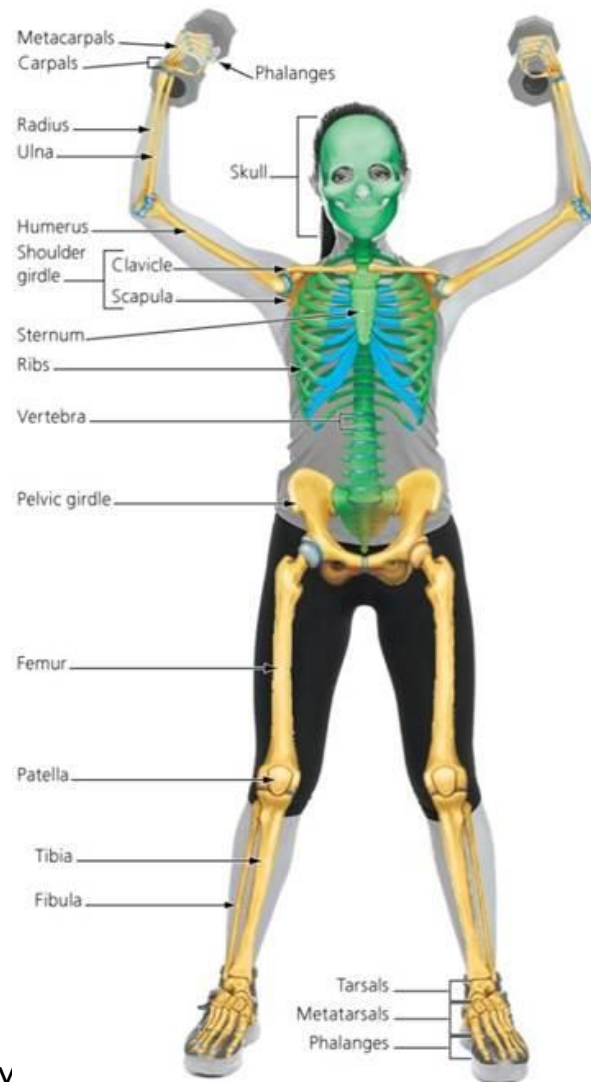
Locomotor Systems

- Movement is one of the most distinctive features of animals and relies upon an interplay of organ systems.
 - Controlled by the nervous system, the muscular system exerts the force that actually makes an animal or its parts move.
 - But the force exerted by muscles produces movement **only when it is applied against a firm structure**.
 - In vertebrates, this structure is the skeletal system.

The Skeletal System: Organization of the Human Skeleton

- The **skeletal system** anchors muscle on the skeleton and provides support and protection of internal organs.
 - All vertebrates have an **endoskeleton**, hard supporting elements situated among soft tissues.
 - The human endoskeleton is a combination of **cartilage and the 206 bones** that make up the skeletal system.

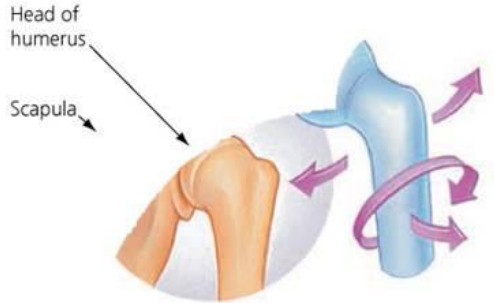
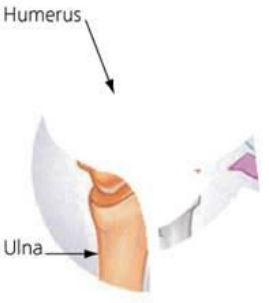
The human endoskeleton



Organization of the Human Skeleton

- Bones are held together at movable joints by strong fibrous tissues called **ligaments**.
- Much of the versatility of our skeleton comes from three types of joint structures.
 1. **Ball-and-socket joints** in the shoulder and hip enable us to rotate our arms and legs and move them in several directions.
 2. **Hinge joints** permit movement in a single direction.
 3. **Pivot joints** enable us to rotate our forearms at the elbow and to turn our heads side to side.

Three kinds of joints

JOINTS		
Ball-and-socket (example: shoulder with movement in several directions)	Hinge (example: elbow with movement in a single direction)	Pivot (example: elbow with partial rotation)
 Head of humerus Scapula	 Humerus Ulna	 Ulna Radius

The Structure of Bones (1 of 2)

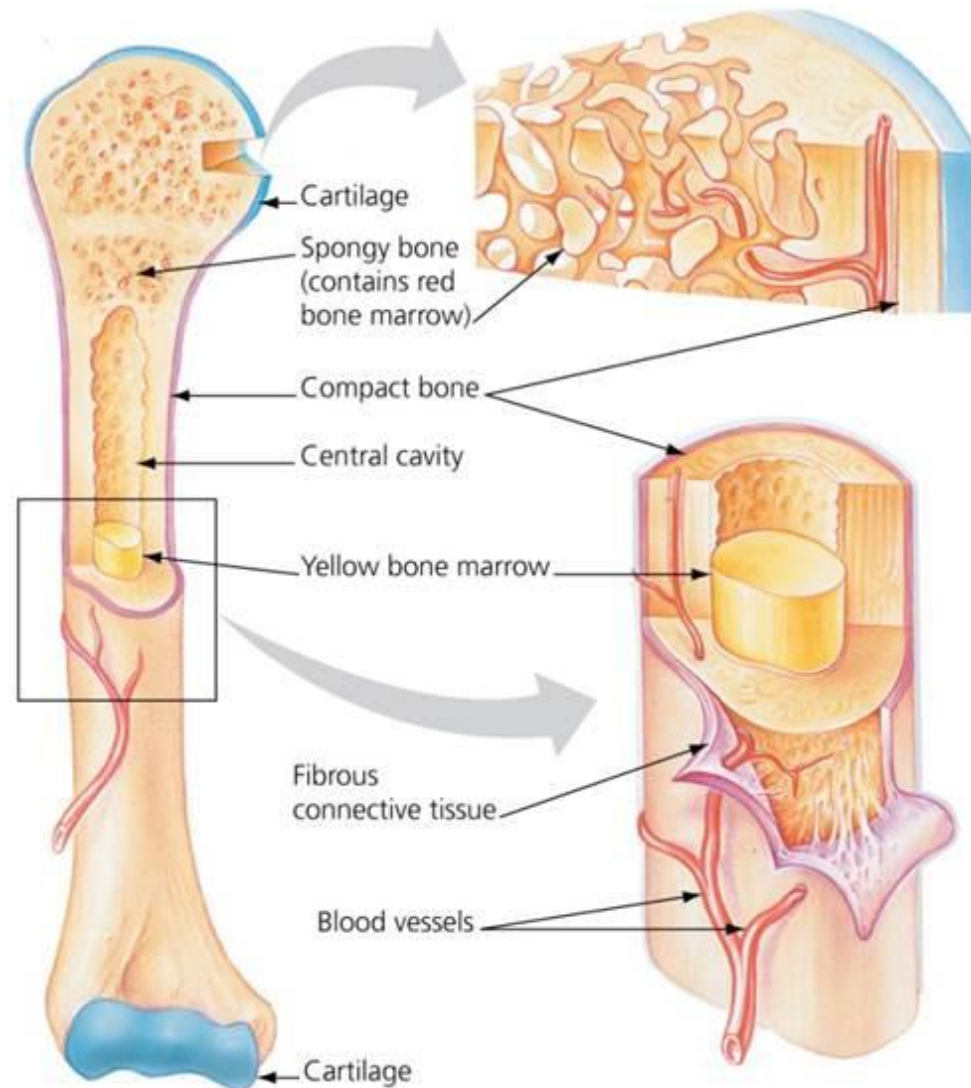
- **Bones** are complex organs consisting of several kinds of living tissues.
 - A sheet of fibrous connective tissue covers most of the outside surface. This tissue can form new bone during normal growth or in the event of a fracture.
 - At either end of the bone is a thin sheet of cartilage which cushions joints, protecting the ends of bones as they move against each other.
 - Bone contains cells that secrete a matrix.
 - Blood vessels course through channels in the bone, transporting nutrients and regulatory hormones to its cells and waste materials from them.

The Structure of Bones (2 of 2)

- The shaft of the long bone surrounds a central cavity. The central cavity contains **yellow bone marrow**, which is mostly stored fat brought into the bone by the blood.
- The ends of the bones have cavities that contain **red bone marrow**, a specialized tissue that produces blood cells.

Shaft: 골간

The structure of an arm bone



Skeletal Diseases and Injuries (1 of 2)

- The skeleton is quite strong and provides reliable support, but is susceptible to disease and injury. If a force is applied that exceeds a bone's capacity to bend, the result is a broken bone, or fracture.
 - The treatment of a fracture involves putting the bone back into its natural shape and immobilizing it until the body's natural bone-building cells can repair the fracture.
 - In more severe cases, a fracture can only be repaired surgically by inserting plates, rods, and/or screws that hold the broken pieces together.

Broken bone



Skeletal Diseases and Injuries (2 of 2)

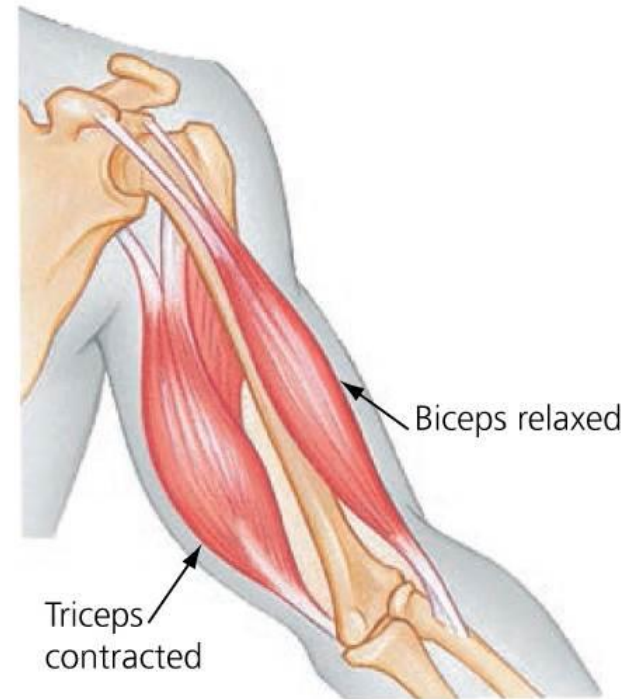
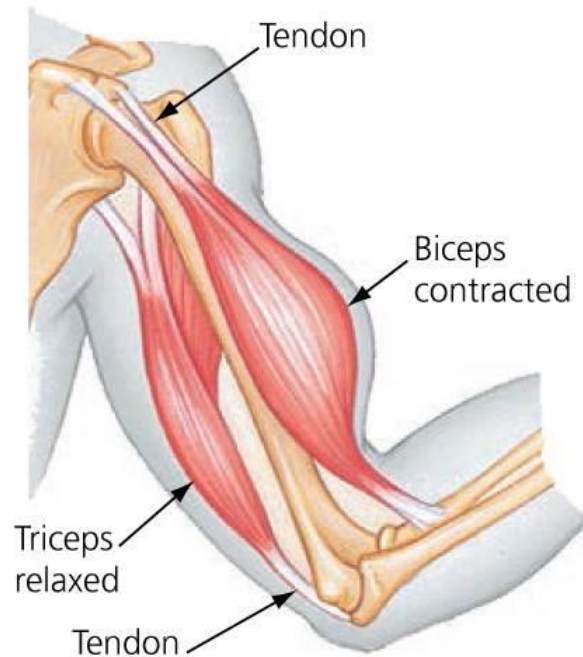
- **Arthritis** is an inflammation of the joints and affects one out of every seven people in the United States.
- **Rheumatoid arthritis** is an autoimmune disease, usually begins between ages 40 and 50, and affects more women than men.
- **Osteoporosis** makes bones thinner, more porous, and more easily broken, and is most common in women after menopause. Insufficient exercise, smoking, diabetes mellitus, and an inadequate intake of protein and calcium may also contribute to the disease.

The Muscular System

- The **muscular system** is made of all the skeletal muscles in the body.
 - **Skeletal muscles** are attached by **tendons** to the skeleton and produce voluntary body movement by interacting with the skeleton.
 - Two other muscle types, **smooth muscle and cardiac muscle**, contract involuntarily and are found inside organs.
 - **Figure 27.31** illustrates how skeletal muscles interact with bones to raise and lower the human forearm.

Tendon: 힘줄, 건

Antagonistic action of muscles in the human arm

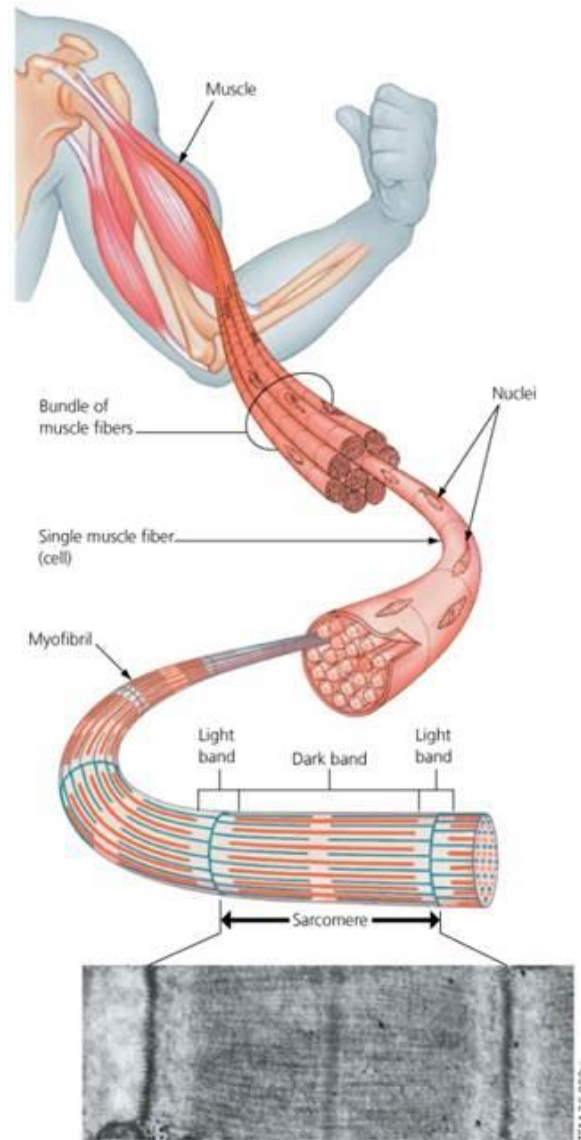


The Cellular Basis of Muscle Contraction (1 of 4)

- A muscle consists of bundles of muscle fibers.
 - Each muscle fiber is a single, long cell with many nuclei and a bundle of smaller **myofibrils**.
 - A myofibril consists of repeating units called **sarcomeres**, the part of the muscle that contracts to make the myofibril shorten.
 - A myofibril is composed of regular arrangements of two kinds of filaments.
 1. **Thick filaments** primarily consist of **myosin**.
 2. **Thin filaments** primarily consist of **actin**.

Myofibril: 근원섬유

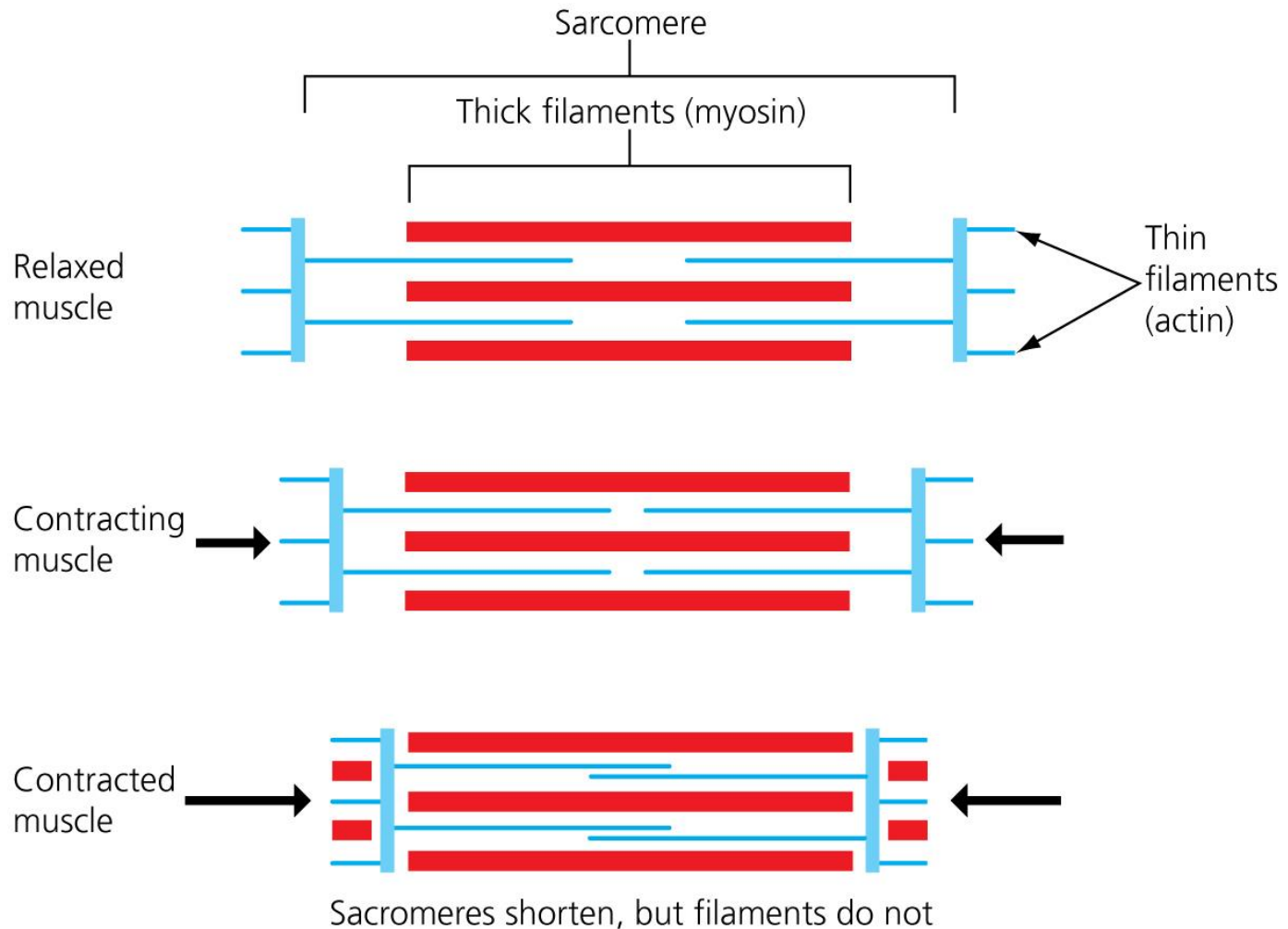
The contractile apparatus of skeletal muscle



The Cellular Basis of Muscle Contraction (2 of 4)

- A sarcomere contracts (shortens) when thin filaments slide past its thick filaments.
 - The model describing this process is known as the sliding-filament model of muscle contraction.
 - **Figure 27.33** is a simplified diagram of this model.

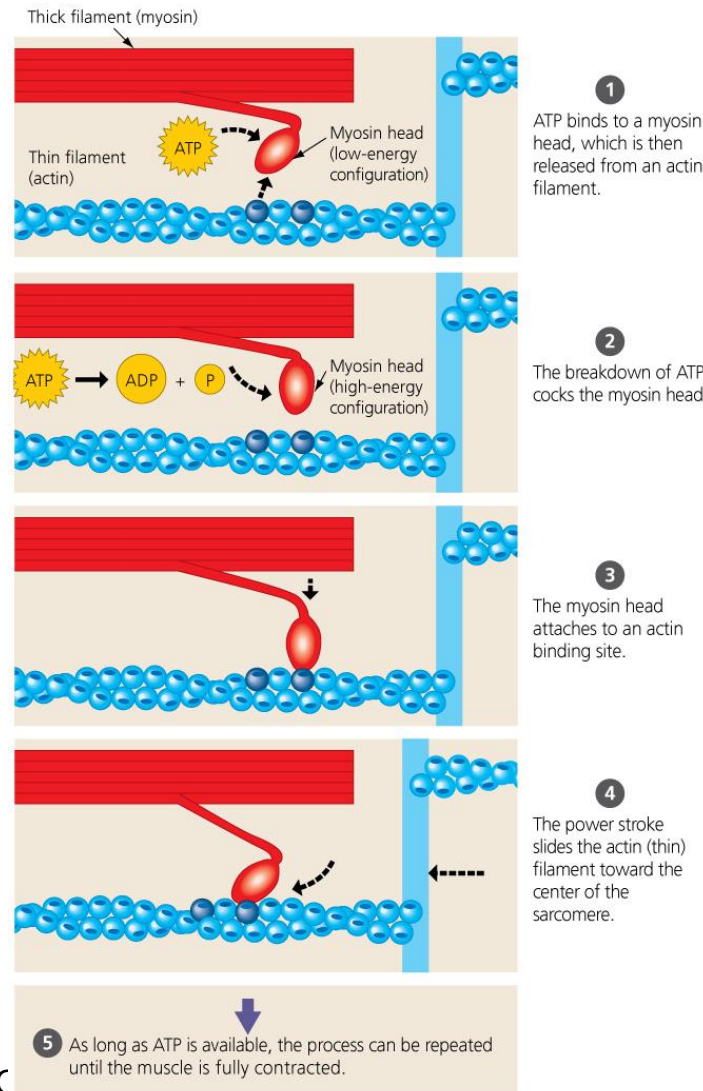
The sliding-filament model of muscle contraction



The Cellular Basis of Muscle Contraction (3 of 4)

- The chemical energy stored in food is transformed to chemical energy in adenosine triphosphate (**ATP**) that your working muscle cells require. Without ATP, thin filaments cannot slide to contract the muscle.
- **Figure 27.34** indicates how sliding is thought to work.

The mechanism of filament sliding



The Cellular Basis of Muscle Contraction (4 of 4)

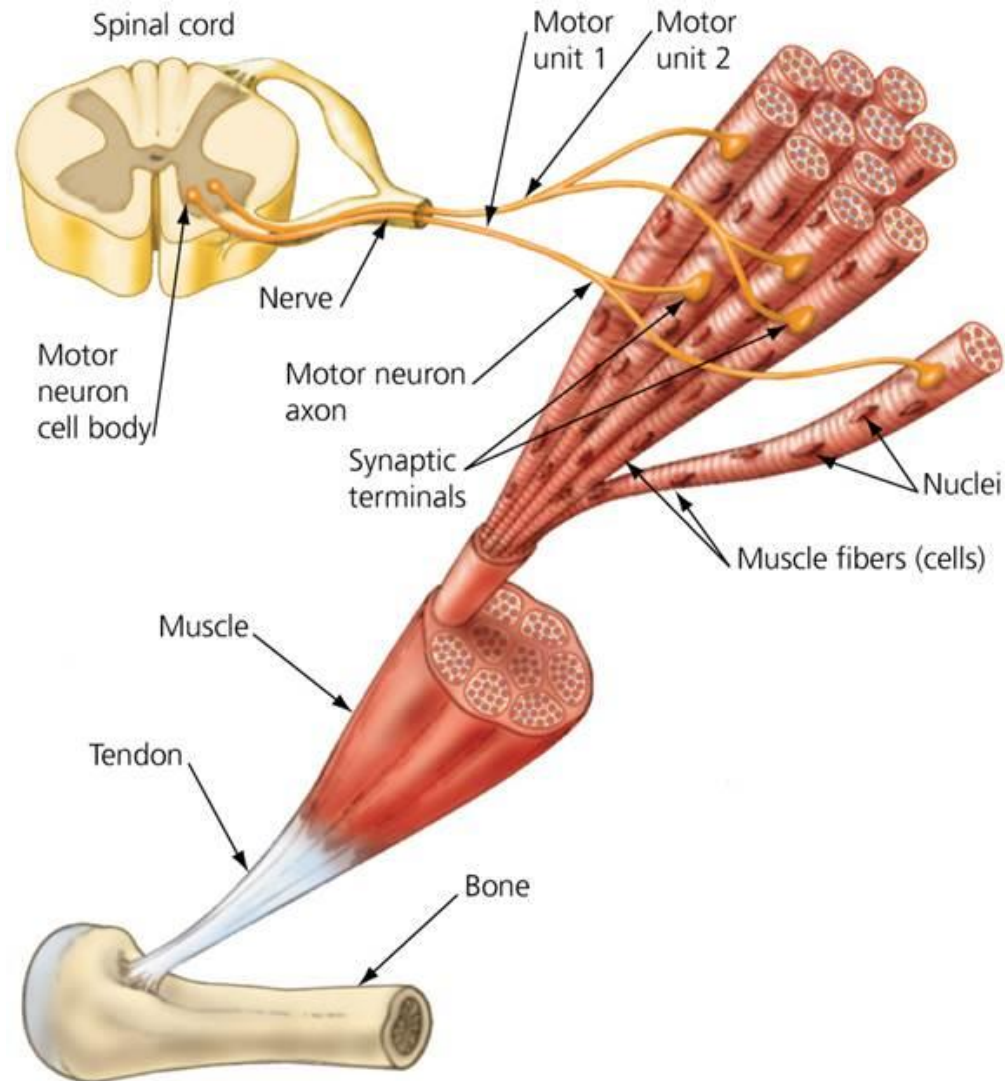
- What if ATP is no longer present, such as when an animal dies? Without ATP, myosin heads are “locked” to actin molecules.
 - The result is rigor mortis (muscle stiffness) that sets in not long after an animal dies.
 - Rigor mortis typically reverses within a few days because as an animal decays, so does its muscle fibers.
 - Since rigor mortis is a temporary condition, forensic scientists examining a dead body can sometimes use the presence of rigor mortis to approximate time of death.

Rigor mortis: 사후 강직

Motor Neurons: Control of Muscle Contraction (1 of 2)

- The sarcomeres of a skeletal muscle fiber are stimulated to contract when a motor neuron releases a neurotransmitter.
 - A typical motor neuron can stimulate more than one muscle fiber because each neuron has many branches.
 - In the example shown in **Figure 27.35**, you can see **two orange motor units**, each consisting of **a neuron** and **all the muscle fibers it controls** (two or three, in this case).

The relationship between motor neurons and muscle fibers



Motor Neurons: Control of Muscle Contraction (2 of 2)

- Each motor neuron may serve just one or up to several hundred fibers scattered throughout a muscle.
 - More forceful contractions result when additional motor units are activated.
 - **Depending on how many motor units** your brain commands to contract, you can apply a small amount of force to lift a fork or considerably more to lift a barbell.
 - In muscles requiring precise control, such as those controlling eye movements, a motor neuron may control only one fiber.

The Process of Science: Can Botulism Toxin Prevent Headaches? (1 of 4)

- **Background:** **Chronic migraines**, which are debilitating headaches accompanied by symptoms such as nausea and sensitivity to light and noise, can make even simple activities agonizing.
 - Over 2% of the population experiences migraines at least 8 days a month and headaches at least 15 days a month.
 - Researchers hypothesized that neurotoxins could help prevent chronic migraines and headaches.

The Process of Science: Can Botulism Toxin Prevent Headaches? (2 of 4)

- **Method:** To test their hypothesis, the researchers recruited almost 700 subjects, people who had a history of migraines.
 - For a month before treatment started, subjects recorded their experiences of migraines and headaches, such as duration and the number of occurrences a day.
 - This information was used as a baseline for evaluating the effectiveness of the experimental treatment that followed.

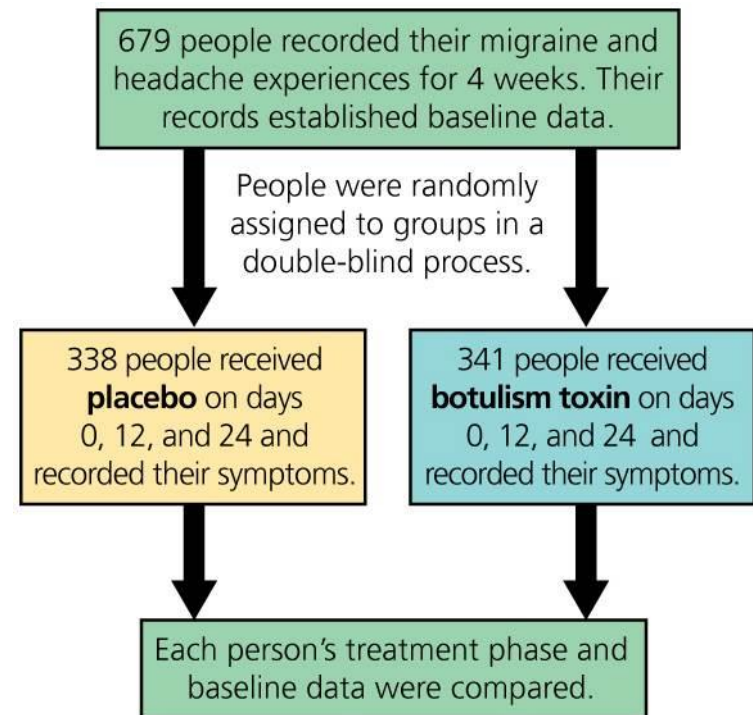
The Process of Science: Can Botulism Toxin Prevent Headaches? (3 of 4)

- Subjects were randomly assigned to either an experimental group or a placebo group.
- On treatment days, the experimental group was injected with botulism toxin (BTX, a neurotoxin) in specific head and neck muscles, and the placebo group participants were injected in the same areas with a saline solution.
- Participants continued to record the same type of data collected during the baseline phase.

Botulism toxin as a potential treatment for headache and migraine



(a) Migraines and headaches can reduce quality of life.



(b) Testing procedure

The Process of Science: Can Botulism Toxin Prevent Headaches? (4 of 4)

- **Results:**

- By week 24, people receiving BTX had about 8 fewer headache days a month.
- People receiving the placebo had about 6.5 fewer headache days a month.
- Although minor, the difference between groups was statistically significant.
- People in the BTX group also reported a significantly higher quality of life. If other studies show that the treatment is safe and effective, neurotoxin may provide hope for chronic migraine sufferers.

Experimental results



(c) Experimental results. Botulism toxin prevented more headache days than the placebo did.

Stimulus and Response: Putting It All Together

- A ball is thrown to a woman and she catches it. This response reflects sensory input, nervous system integration, and locomotor responses.
 - The woman obtains information with sensory structures. Photoreceptors in her eyes track the ball.
 - Sensory neurons convey this information to her brain as a series of action potentials (sensory input).
 - Within her brain, information about the angle and speed of the ball is interpreted (integration).
 - In response, motor neurons signal multiple muscles to perform very specific actions (motor output).

The nervous, sensory, and locomotor systems in action



Evolution Connection: A Neurotoxin Arms Race (1 of 2)

- Some prey use neurotoxins as protection against predators. The rough-skinned newt produces a toxin that can potentially paralyze its predators, such as garter snakes, which are not poisonous.
 - The newt's neurotoxin must be potent enough to slow and weaken the snake to allow the newt time to escape the snake.
 - The snake with a mutation in a gene that results in a resistance to the newt's neurotoxin, would have an advantage over snakes without that mutation.

Newt: 영원 (도룡뇽목 영원과의 동물)

Evolution Connection: A Neurotoxin Arms Race (2 of 2)

- Over many generations of random mutation and nonrandom selection, rough-skinned newts with **greater and greater neurotoxicity** were selected for, while garter snakes with **greater and greater resistance** to that neurotoxin were selected for.
- Evidence for the evolution of variation in toxicity and resistance to toxicity can be seen in various locations along the west coast of North America.
- Such a predator-prey “arms race” is a common outcome of evolution by natural selection and a potent reminder of the importance of the nervous system to animal form and function.

The consumption of a rough-skinned newt by a garter snake



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